

Understanding active fire detection uncertainty with Bayesian Neural Networks

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Abstract:

Active fire data from public and civilian satellites are used all over the world to inform wildfire management and decision-making. However, falsely detected wildfires can impact the quality and reliability of the data while also propagating errors into subsequent analyses. In this chapter, we present a tutorial-style guide that demonstrates how Bayesian Neural Networks (BNNs) can be used to classify active fire detections as true or false positives, while also quantifying the uncertainty within each prediction. This probabilistic approach is intended to provide additional context around individual detections for users of satellite-based fire monitoring systems. We use a case study of Canada's record-breaking 2023 fire season to consider two methodologies: (1) pre-screening VIIRS active fire data to identify and remove likely false positives, and (2) assessing detection confidence across multiple Earth observation sources (MODIS and VIIRS). We evaluate model performance under varying data processing and hyperparameter tuning strategies, with the goal of offering a transparent, adaptable workflow for future studies and possible integration into operational fire detection systems. While the results are exploratory in nature, this chapter is intended as a foundation for researchers interested in uncertainty-aware fire detections using BNNs. All code, including a complete workflow and supporting functions, is provided in the jupyter lab and Google Colab notebooks to facilitate adaptation to other regions, datasets, or time periods.

Keywords:

Bayesian Neural Networks; fire monitoring; active fire hotspots; MODIS; VIIRS; wildfire detection; modelling uncertainty

1. Introduction

Wildfires are a natural Earth process that is part of many landscapes and ecosystems, as exemplified by the role of fire in Boreal forest succession found in many parts of Canada and other high-latitude regions around the world (Girardin et al., 2010; Girardin & Mudelsee, 2008). However, due to a changing climate, fire seasons are lengthening in duration, and the severity and variability of wildfires are increasing in Canada and worldwide (Cunningham et al., 2024; Jones et al., 2022; Skakun et al., 2024; W. Wang et al., 2025; X. Wang et al., 2017). Canada's 2023 wildfire season set records for area burned across multiple provinces and territories (Beverly & Schroeder, 2025; Boulanger et al., 2025; Daniels et al., 2024; Jain et al., 2024), a trend exacerbated by anthropogenic climate change (Kirchmeier-Young et al., 2024). Extreme or unpredictable wildfires can lead to loss of life and cause long-term damage to ecosystems, biodiversity, and critical infrastructure (Coogan et al., 2021; Tymstra et al., 2020). There have been calls for novel fire monitoring methods to inform effective decision-making strategies and reduce the negative while promoting the positive impacts of wildfires (Crowley et al., 2023).

Wildfire detection and mapping is a rapidly evolving field, with extensive research and development focused on advancing data collection methods to meet diverse user needs, and each approach offers distinct advantages and limitations (Chen et al., 2024; Chuvieco et al., 2019, 2020; Crowley et al., 2023; Johnston et al., 2020; San-Miguel-Ayanz & Ravail, 2005; Wooster et al., 2021). Ground-based observations and RPAS (Remotely Piloted Aircraft System) are useful for on-demand active fire monitoring by providing high-resolution data in targeted areas; however, they are limited by spatial coverage, rendering them challenging for monitoring large or cross-boundary fires (Crowley et al., 2023). Satellite-based monitoring has become an important component of operational fire management strategies for wildfire detection and mapping, due to its large spatial coverage and high temporal resolution (Crowley et al., 2023; Wooster et al., 2021). In Canada, final fire perimeters created at the end of the fire season (i.e., the boundaries of burning or burned areas) are typically derived from a combination of methods, using satellite data where available and supplementing with ground or aerial-based agency reports when necessary (Skakun et al., 2022). Satellite data also plays a crucial role in long-term monitoring and trends in wildfire activity (Skakun et al., 2024), which is important for assessing the impacts of climate and creating informed mitigation strategies (Chuvieco et al., 2019; Crowley et al., 2023).

Several government organizations operating Earth observation (EO) satellites offer publicly available data, promoting accessibility and collaboration among researchers, organizations, and governments (Crowley et al., In Review). While there are currently no government-owned and operated satellites in orbit designed specifically for monitoring active wildfires as the primary function, Canada will launch WildFireSat in 2029, which has been designed for and with operational fire managers (Johnston et al., 2020; McFayden et al., 2023b; Thompson et al., 2025). Although not specifically developed for fire monitoring, several EO satellites currently in orbit are equipped with instruments that can be used to detect fires and monitor environmental changes (Crowley et al., 2023; Johnston et al., 2020; Wooster et al., 2021). Key examples include the Visible Infrared Imaging Radiometer Suite (VIIRS) sensor on the NASA (National Aeronautics and Space Administration) Suomi NPP (National Polar-orbiting Partnership) and the NOAA (National Oceanic and Atmospheric Administration) 20/21 satellites, the MODIS (Moderate Resolution Imaging Spectroradiometer) sensor on NASA's Terra and

Aqua satellites, the ABI (Advanced Baseline Imager) sensor on NOAA's GOES (Geostationary Operational Environmental Satellites) series satellites, and the OLI/OLI-2 (Operational Land Imager) and TIRS (Thermal Infrared Sensor) sensors on the Landsat series satellites. Additional satellites from the European Space Agency's Copernicus mission provide a variety of fire monitoring information, including physical changes from Sentinel-1 C-Band SAR (Synthetic Aperture Radar), post-fire information from Sentinel-2 MSI (Multispectral Imager), active fire data from Sentinel-3 SLSTR (Sea and Land Surface Temperature Radiometer) and the MTG (Meteosat Third Generation) constellation, and emissions from Sentinel-5P (Precursor) Tropomi sensor. Government-funded web platforms such as NASA FIRMS (Fire Information for Resource Management System; <https://firms.modaps.eosdis.nasa.gov/>), GWIS (Global Wildfire Information System; <https://gwis.jrc.ec.europa.eu/>), and INPE's Queimadas (Instituto Nacional de Pesquisas Espaciais; <https://terrabrasilis.dpi.inpe.br/queimadas/portal/>), are frequently used by operational fire managers and the general public around the world to access fire monitoring information in near-real-time (D. K. Davies et al., 2015; Hope et al., 2024).

The detection of actively burning wildfires using satellite imagery is typically supported by algorithms designed to identify pixels in the images that contain radiative energy above a certain threshold (Giglio et al., 2016; Schroeder et al., 2014; Wooster et al., 2021). The identified high energy pixels are commonly referred to as thermal hotspots. These algorithms can be run in near real-time, allowing for the detection of hotspots as soon as data becomes available. Hotspot data is included in publicly available active fire products, which typically provide the latitude and longitude of the fire pixel's center, along with a defined resolution that varies by instrument. The accuracy of hotspot placement is influenced not only by the spatial resolution of the satellite but also by factors such as satellite viewing geometry (azimuth and elevation), geolocation accuracy, orthorectification, and the sensitivity of the sensors wavebands (Johnston et al., 2018; Schroeder et al., 2014). These combined factors can introduce uncertainty in the precise location of detected fire activity. Additionally, because fires tend to burn hotter in the late afternoon to early evening, satellite overpasses that miss this period may fail to detect newly ignited fires (Johnston et al., 2020). While hotspots are frequently used as one of many approaches for operational fire monitoring, errors and limitations can reduce user trust in the satellite data when not properly communicated or quantified (Crowley et al., 2023b; McFayden et al., 2025, 2023b; Thompson et al., 2025).

Bias in machine or deep learning model outputs can arise from various sources, including systematic errors due to incorrect assumptions, biases in training data, or limitations in model architecture and learning objectives (Mavrogiorgos et al., 2024). In regards to satellite-based fire monitoring this bias commonly manifests as the frequent persistent occurrence of false positives, driven by both known and unknown causes. False positives can occur due to issues like incorrect pixel projections near the edge of the satellite swath or due to superheated smoke plumes drifting outside the active fire area (Earth Science Data Systems, 2024a; Schroeder et al., 2014). Another source of a false positive is sunglint, or the misidentification of light being reflected off water bodies, shores, and bright areas (Crowley et al., 2023; Giglio et al., 2016; J. M. Johnston et al., 2020). Other false positives can be due to persistent heat sources misclassified as fire anomalies, such as from industrial plants or volcanoes, as have been documented in datasets developed by NASA FIRMS, Natural

Resources Canada, and the upcoming WildFireSat mission (D. Davies, 2025; Johnston et al., 2020; Thompson et al., 2025).

Another potential source of error is the misinterpretation of multisource datasets. Active fire detection algorithms typically produce probabilistic indicators, not strictly binary results, representing the likelihood of an active fire at a given location (Schroeder et al., 2014). When algorithm output exceeds a set threshold, a pixel is flagged as an active fire and assigned a confidence or probability value. Higher confidence levels are thought to be associated with fewer false positives, while lower confidence values can carry greater uncertainty. Additionally, each EO sensor (such as MODIS or VIIRS), employs its own detection algorithm, thresholds, and confidence classifications. As a result, these products exhibit differing rates of uncertainty and false positives (Fu et al., 2020). When integrating multisource datasets, it is essential to consider how introducing data from a sensor with the potential of false positive detections could affect downstream modelling and interpretation. If not properly identified, errors in active fire data can be perpetuated in subsequent decision-making and analyses for fire managers and other users (Crowley et al., 2023; Johnston et al., 2020; McFayden et al., 2023b). These challenges can lead to confusion in resource allocation and delays in responses (Crowley et al., 2023b; McFayden et al., 2025). To mitigate these impacts, data producers should provide robust uncertainty metrics that users can incorporate into their workflows, enabling more informed decisions about whether to act on a detected hotspot (Crowley et al., In Review; McFayden et al., 2023a).

Recent advancements in technology and artificial intelligence (AI) have shown promise for improving the automation and precision of wildfire monitoring using satellite data. A growing body of research highlights the effectiveness of deep learning algorithms for detecting smoke and fire, demonstrating high model performance (Ba et al., 2019; Ghali & Akhloufi, 2023; Gonçalves et al., 2024; Larsen et al., 2021; Liu et al., 2024). Additionally, deep learning has been successfully applied to burned area mapping (Alencar et al., 2022, 2022; Ban et al., 2020; Cho et al., 2023; Hu et al., 2021; Knopp et al., 2020; Seydi et al., 2022; Zhao & Ban, 2025), with these methods showing strong agreement when compared to historical fire perimeters. Building upon these advances, there is potential for AI-based models to be used to enhance traditional methods of bias reduction when identifying errors and quantifying uncertainty in existing active fire datasets. Specifically, the availability of large amounts of input data and auxiliary data sets that can be used for validation, alongside well-documented cases of error, makes fire monitoring using EO data a classification problem well-suited for AI-driven approaches, especially deep learning models trained to flag and interpret anomalies.

1.1 Objective

This chapter explores the application of Deep Learning (DL), with a focus on Bayesian Neural Networks (BNNs), for fire monitoring using satellite-detected hotspots. Some BNN architectures incorporate uncertainty into their predictions, which can allow for estimating confidence intervals or predictive distributions. This added layer of information can improve transparency and better support critical decision-making by stakeholders. This chapter focuses on outlining the technical considerations and workflow necessary for data retrieval, preparation, model design, training and evaluation of our BNN. This framework ensures readers can apply this knowledge in developing their own DL solutions, rather than providing a highly tailored

model with high performance for this specific application. The accompanying code is written in Python and provided as a Google Colab notebook. The notebook includes all instructions required to run the code. Users are expected to have a basic familiarity with Python and Google Colab, including loading data into the notebook and executing code cells. Users who are not already familiar with these tools are encouraged to review publicly available Google Colab tutorials before working with the notebook.

This tutorial is heavily influenced by the methodology presented in Keras' official Probabilistic Bayesian Neural Networks tutorial first published in 2021 (Keras, 2021). It is not a requirement to be familiar with this tutorial to complete the one that accompanies this chapter. The goal of this project is to add additional value to the existing Keras tutorial by including additional model performance evaluation and improvement strategies. A trained model must undergo additional evaluations to ensure it is performing as the user requires. Included in our workflow are some techniques that can be used to perform these evaluations as well as directions for how to interpret the results of these evaluations. To illustrate this approach, we present two example case studies (detailed further in Section 3.3) that are relevant to fire monitoring in Canada:

- 1) **Case Study A:** Pre-screening VIIRS active fire data to highlight potential false positives using BNN. This study uses a point prediction BNN model, which outputs a single probability score for each detection, indicating the likelihood it is a true positive. Applying a classification threshold to this score yields a binary indicator (true/false positive), primarily for use in pre-screening VIIRS data to identify and remove likely false positives.
- 2) **Case Study B:** Assigning multi-source EO active fire confidence using a probabilistic BNN. This study uses a probabilistic BNN architecture that outputs a full probability distribution for each detection, not just a single point estimate. The resulting output includes a quantified measure of uncertainty (the standard deviation), which allows for the calculation of a confidence interval around the predicted probability. Furthermore, this case study integrates data from multiple Earth Observation sources (MODIS and VIIRS) to assess the impact of sensor differences on classification uncertainty.

2. Study Area and Data

2.1 Study Area

This study focuses on the 2023 wildfire season, which was the most severe in Canada's recorded history, with over 15 million hectares burned (Jain et al., 2024; Kirchmeier-Young et al., 2024). The unprecedented scale and intensity of these fires were driven by compounding factors such as early snowmelt, prolonged drought, and anthropogenic climate change (Beverly & Schroeder, 2025; Boulanger et al., 2025; Daniels et al., 2024; Jain et al., 2024). To reduce computational load for the purposes of this tutorial, we focus on a single representative day: September 23rd, 2023. This date falls near the end of the 2023 fire season and captures a period when large, persistent fires were active across multiple provinces and territories. By analyzing nationwide fire detections on this day, the study evaluates satellite-based hotspot monitoring across diverse ecological zones and fire conditions, supporting the assessment of detection consistency, uncertainty, and spatial distribution at a national scale.

2.2 Data

For active fire monitoring during the fire season using satellite-derived hotspots, the Canadian Forest Service (CFS) has historically relied on data from the AVHRR (Advanced Very High Resolution Radiometer), MODIS and VIIRS instruments (Canada, n.d.; Fraser et al., 2000). AVHRR is no longer in regular use by CFS, and MODIS is nearing the end of its operational life (de Jong & McFayden, 2023), therefore, users have been encouraged to transition to VIIRS, which offers comparable products with enhanced detection capabilities. Accordingly, this study focuses primarily on active fire data derived from the VIIRS sensor. Active fire data are obtained from FIRMS (Earth Science Data Systems, 2024b), which distributes many products including the VIIRS 375 m active fire product. FIRMS provides four data quality levels: ultra real-time, near-real-time, real-time, and standard quality. While real-time data are valuable for early detection, the reported hotspot locations may be less accurate. Standard-quality data undergo additional processing and calibration to improve geolocation and reliability. To ensure positional accuracy, only standard-quality data are used in this study, corresponding to all available data from the NOAA-20 and Suomi NPP satellite overpasses. At the time of analysis, standard quality data from NOAA-21 were not available for the time period and region of interest, and as a result, NOAA-21 data were excluded from further analysis. The available input features within the VIIRS dataset are displayed in Table 1. To subset FIRMS hotspots to Canada only, a shapefile of Canada is obtained from Natural Earth (Table 2). Although the Government of Canada provides an official country boundary shapefile, we are using Natural Earth shapefiles in this study to demonstrate where users can access perimeter data for multiple countries (see Table 2).

Certain areas of consistent thermal activity, such as human infrastructure like power plants, are frequently misidentified as active fires by fire detection products. A shapefile of these persistent hotspots comes from an internal source at Natural Resources Canada and is used to exclude them from the hotspots of interest.

Fire perimeter data are obtained from the National Burned Area Composite (NBAC) (Natural Resources Canada, 2025b) and the Canadian National Fire Database (CNFDB) (Natural Resources Canada, 2025a), which collectively represent the best available estimate of burned area within Canada. Figure 1 displays all NBAC, CNFDB, and persistent hotspot perimeters used in this tutorial.

Table 1: Features available through the NASA FIRMS VIIRS 375 m active fire product.

Inputs	Short Description	Long Description
Latitude	Latitude	Center of pixel.
Longitude	Longitude	Center of pixel.
bright_ti4	Brightness temperature I-4	Brightness temperature (Kelvin) of the hotspot pixel for VIIRS I-4 channel.
Scan	Along scan pixel size	Actual pixel size.
Track	Along track pixel size	Actual pixel size.

acq_time	Acquisition Time	Time of acquisition of the satellite.
satellite	Satellite	For both case studies, false positives are represented by class 0 and true positives are represented by class 1. For Case Study B, source 0 is VIIRS and source 1 is MODIS.
confidence	Confidence	For Case Study A, low confidence is mapped to 0, nominal confidence is mapped to 1, and high confidence is mapped to 2. For Case Study B, VIIRS confidence values are mapped to 33% for low, 66% for nominal, and 100% for high (MODIS confidence scores are inherently percentages).
bright_ti5	Brightness temperature I-5	Brightness temperature (Kelvin) of the hotspot pixel for VIIRS I-5 channel.
frp	Fire radiative power	Pixel-integrated fire radiative power in megawatts (MW).
daynight	Acquisition time	D = Day acquisition, N = Night acquisition.

Table 2: Data sources used in this tutorial and their links. Additional links are provided for users to reference, as they contain valuable information that was used to develop this tutorial.

Data Source	Link
NBAC	https://cwfis.cfs.nrcan.gc.ca/datamart/metadata/nbac
CNFDB	https://cwfis.cfs.nrcan.gc.ca/ha/nfdb
CWFIS	https://cwfis.cfs.nrcan.gc.ca/home
FIRMS	https://firms.modaps.eosdis.nasa.gov/map
FIRMS Tutorials	https://firms.modaps.eosdis.nasa.gov/academy
FIRMS FAQ	https://www.earthdata.nasa.gov/data/tools/firms/faq
Natural Earth	https://www.naturalearthdata.com/

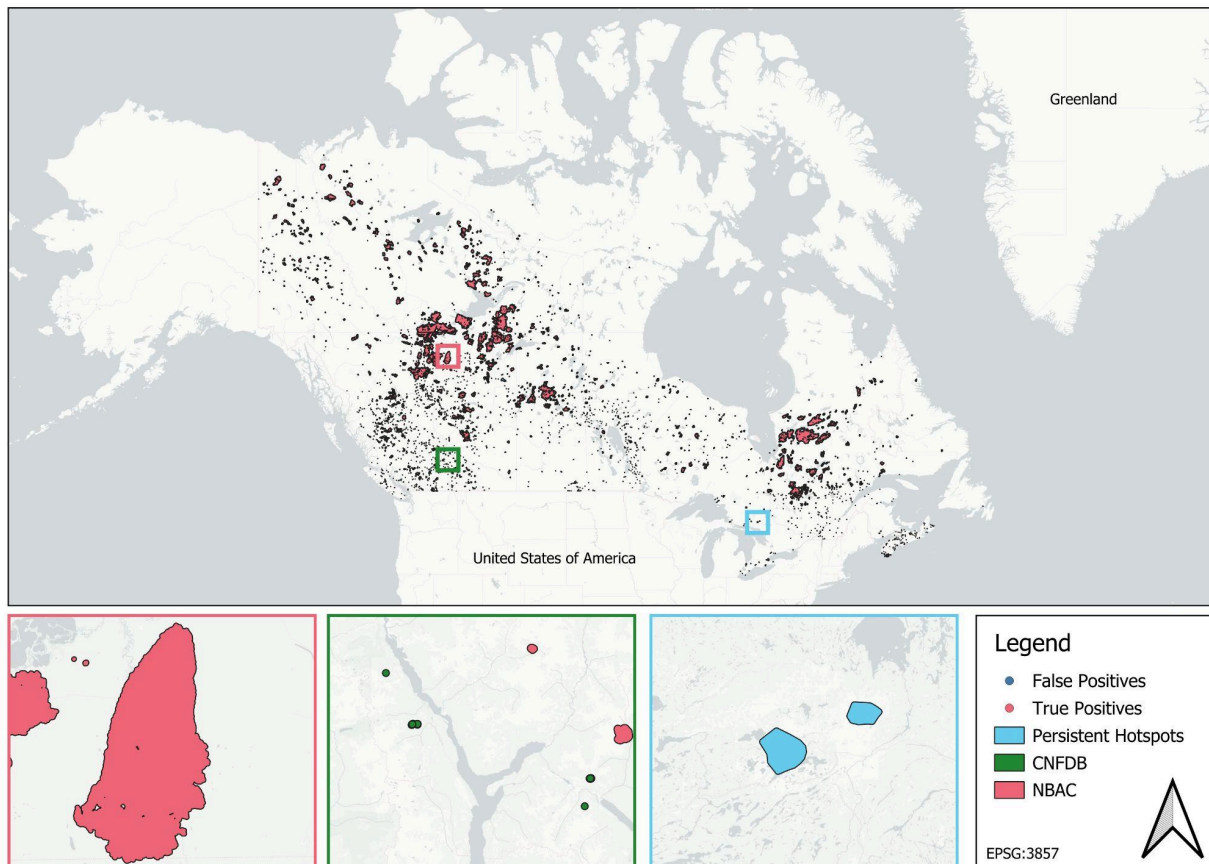


Figure 1: Map of all known fire perimeters for 2023 in Canada. These perimeters are all buffered by 750 m, as explained in the data cleaning and preprocessing section. The three panels at the bottom display zoomed in examples of each perimeter type, with NBAC (National Burned Area) in red, CNFDB (Canadian National Fire Database) in green, and persistent hotspots in blue.

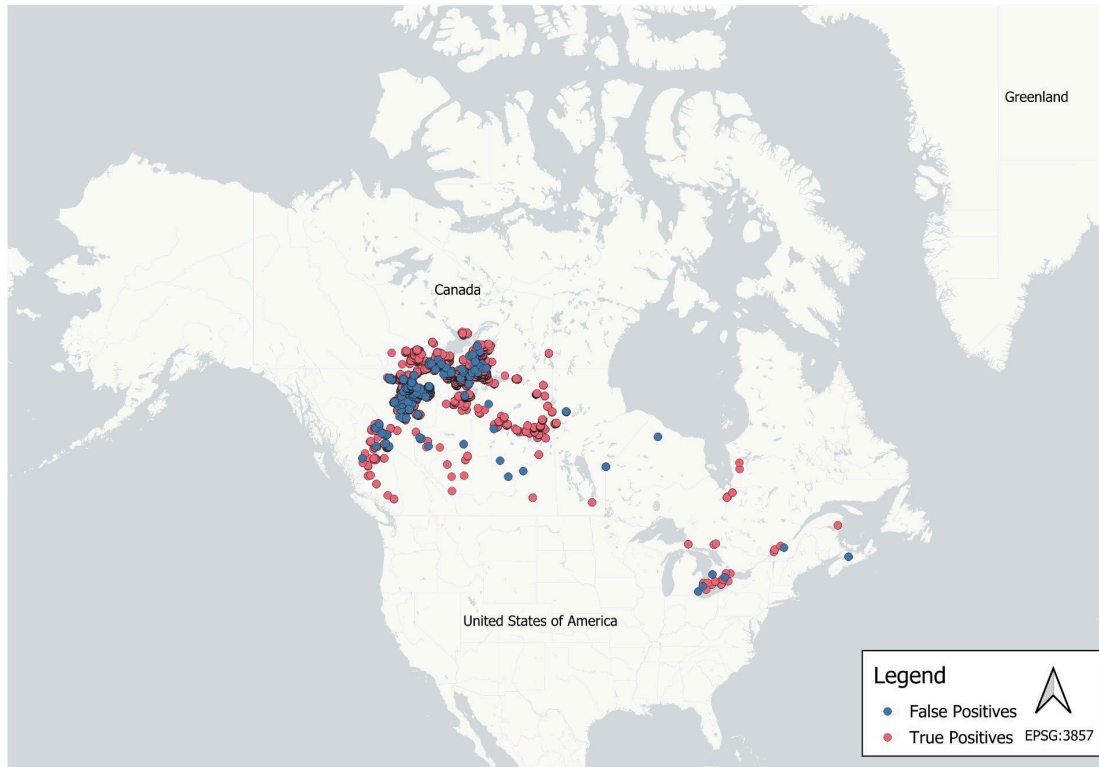


Figure 2: Map of all hotspots detected by the VIIRS 375 m active fire product on the date of interest (September 23rd, 2023) within Canada. True positives are defined as hotspots within known perimeters, and false positives are defined as hotspots outside of known perimeters.

3. Background and Methods

In this section, we discuss some deep learning (DL) basics, including a few different types of architectures, the structure of BNNs, and our model implementation and evaluation techniques.

3.1. Deep Learning Basics

The application of deep learning in the field of environmental sciences is progressing rapidly, especially in the field of wildfire (Jain et al., 2020). When beginning a DL-based project, it is essential to carefully consider the type of model architecture that would be most appropriate for the task at hand. For our project, we limit our model selection scope to DL-based methods; more specifically, we focus on different types of artificial neural network (ANN) architectures. For some projects, more classic algorithms such as a support vector machine or random forest can result in very high model performance, and so it is often worth considering these more “basic” approaches alongside more involved ANN methods. Artificial neural networks (ANNs) come in a variety of architectures, from simpler models like the perceptron and basic feedforward neural networks (FNNs), to more complex designs such as convolutional neural networks (CNNs) (O’Shea & Nash, 2015) and recurrent neural networks (RNNs) (Schmidt, 2019).

The typical structure of a basic ANN contains three distinct layer types: an input layer, a hidden layer(s) and an output layer(s) (Kufel et al., 2023; Panchal et al., 2011; Wu & Feng,

2018). The structure of the input and the output layers are vectors whose function is to accept information and output information respectively. Hidden layers in ANNs consist of learnable parameters such as weight and biases, combined with activation functions (Panchal et al., 2011; Wu & Feng, 2018). In more complicated ANN architectures, hidden layers may also include additional components such as convolutional kernels, recurrent units, attention mechanisms, or normalization layers. In all types of ANNs, during model training the coefficients are adjusted, typically by implementing backpropagation, to minimize the total model error (Kufel et al., 2023; Wu & Feng, 2018). Training an ANN with a single hidden layer is called single layer ML or shallow NN, and training an ANN with multiple hidden layers is called multiple layer ML or a deep neural network (DNN) (Kufel et al., 2023). Examples of shallow and deep ANN architectures are displayed in Figure 3. Typically, ANNs with more than one hidden layer tend to outperform single layer models, and we use this knowledge to our advantage and we focus on developing a DNN (Kufel et al., 2023).

By default, most DNNs are deterministic, meaning they produce the same output given the same input and parameters (Lee et al., 2016). This can be appropriate for problems with well-understood relationships and low uncertainty. However, many real-world problems, such as active fire detection using Earth observation data, involve significant uncertainty due to sensor limitations, atmospheric interference, and the stochastic nature of fire behaviour. In such cases, a probabilistic or stochastic modeling approach is often more appropriate. A DNN can be enhanced to include classification uncertainty by incorporating probabilistic elements, such as Bayesian layers, effectively turning them into stochastic models capable of expressing uncertainty in their predictions (Mullachery et al., 2018).

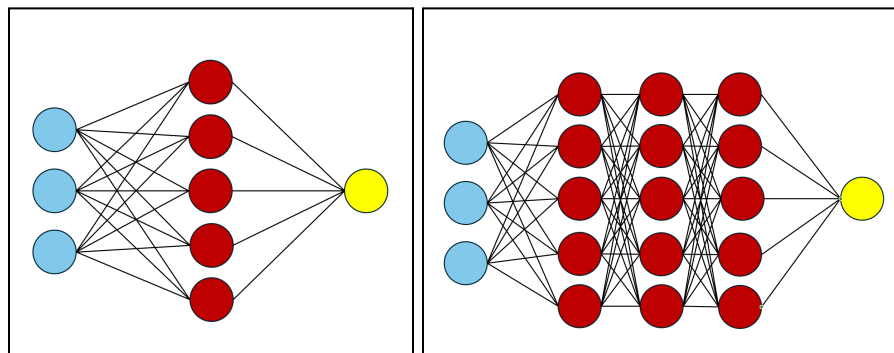


Figure 3. Visualization of the structure of shallow (left panel) and deep (right panel) ANN architectures. Blue circles represent nodes within the input layer, yellow the output layer, and red the hidden layer(s). Note that the output layer can also have multiple nodes.

In this tutorial, we are working with active fire data pulled from the FIRMS archive (as discussed in Section 2.2) which is in tabular format. In general, tabular data requires less memory to store, has quicker processing times, and is typically considered easier to work with than image or sequence data. We therefore chose to not utilize CNNs nor RNNs, as they require different input data formatting as well as additional preprocessing for non-sequential tabular data. As our aim is to keep our tutorial as accessible and streamlined as possible, we focus on models that are more suitable to the format of our input data.

3.2. Bayesian Neural Networks

As discussed previously, our goal is to implement a model architecture that can accept FIRMS data in tabular format and be configured to incorporate uncertainty into its predictions. To meet these requirements, we selected BNNs as our modeling approach. A multilayer BNN is a type of deep neural network in which the model parameters (specifically, the weights and biases) are represented as probability distributions rather than fixed values (Blundell et al., 2015). The inclusion of distributions with model architecture gives the model the potential to capture uncertainty in its predictions. Using Bayes' theorem, the model updates these distributions as it observes new data (Arbel et al., 2023; Mullachery et al., 2018). The prior distribution for the weights is a tunable hyperparameter that can be set to reflect prior knowledge or assumptions.

In a BNN, each hidden layer still includes a weight matrix (kernel), bias vector, and activation function. However, these parameters are sampled from learned distributions rather than being optimized as single point estimates. The final layer is typically a dense layer with a Sigmoid activation function, producing a 1D vector of probabilities, one for each input sample, representing the model's estimated probability that an input belongs to class 1. Values closer to 1 indicate higher confidence in class 1 and values closer to 0 suggest class 0.

A specific subtype of BNN, the probabilistic BNN, takes this further by modeling the output as a full probability distribution rather than a single predicted probability. The shape of this output distribution is a configurable hyperparameter in the model architecture. To interpret these outputs, the mean of the distribution is used as the predicted probability. The standard deviation of the distribution reflects the model's confidence in that prediction, and a higher standard deviation indicates greater uncertainty. This enables the calculation of confidence intervals, which provide a range around the predicted probability and act as an uncertainty measure. Such information is particularly valuable in high-risk decision-making contexts. In this setup, the final Sigmoid layer is replaced by an Independent Bernoulli layer (Keras, 2021), which models each output unit as an independent Bernoulli-distributed random variable. This layer outputs a probability distribution over binary outcomes rather than fixed values, making it suitable for probabilistic binary or multi-label classification tasks. The mean of the output distribution represents the predicted probability, while its standard deviation quantifies the uncertainty associated with that prediction.

3.3 Model Architecture and Implementation

In this project, we define, train and evaluate four standard BNNs and one probabilistic BNN. They are implemented using the deep learning API Keras (v2.2) built on top of a TensorFlow (v2.2.0) backend framework (Abadi et al., 2016). All code for this project is written in Python 3 (version 3.10) (Python, 2021) and hosted in a Google Colab (Google, 2025) notebook. Our methodology is heavily influenced by the methodology presented in Keras' official Probabilistic Bayesian Neural Networks tutorial first published in 2021 (Keras, 2021). The following sections describe the architectures of two distinct Bayesian Neural Network (BNN) models developed for this tutorial. Both models use probabilistic layers provided by TensorFlow Probability (Tensorflow, 2025). Figure 4 presents the base architecture shared by all BNN

models in this tutorial. Sections 3.3.1 and 3.3.2 describe the model architectures used in more detail.

3.3.1. Point Prediction BNN

The first BNN we develop outputs a single point probability prediction, while applying Bayesian principles within its hidden layers. The model begins by concatenating all individual inputs into a single feature vector. This combined vector passes through a batch normalization layer, which standardizes (re-centers and re-scales) the features to improve training stability and convergence.

Next, the vector is processed by Bayesian hidden layers, implemented using TensorFlow Probability's "DenseVariational" layers (Tensorflow, 2025). These layers are analogous to standard dense (fully connected) layers but treat their weights and biases as probability distributions rather than fixed parameters (Blundell et al., 2015). Each layer learns both a mean and variance for these parameters, allowing the model to represent uncertainty in its internal representations. We use two DenseVariational layers with eight hidden units each, and a rectified linear unit (ReLU) activation function.

The prior distribution defines the initial belief about the weights before observing any data. We use a fixed, non-trainable TensorFlow MultivariateNormalDiag layer (Tensorflow, 2025) with zero mean and unit variance, representing a standard normal prior. This choice assumes that weights are initially independent and normally distributed.

The posterior distribution represents what the model learns about the possible values of its internal parameters after seeing the data. We model this using a trainable MultivariateNormalTriL layer (Tensorflow, 2025), which allows the model to learn not only the mean and variance of these parameters, but also how they may be related to one another.

As was done in the Keras tutorial (Keras, 2021), we use the Kullback-Leibler (KL) divergence penalty (Shlens, 2014) which measures how much the learned posterior distribution over the network weights diverges from the prior distribution. In a Bayesian framework, minimizing this divergence encourages the posterior to stay close to the prior unless the data provide strong evidence otherwise. This acts as a form of regularization, preventing overfitting and ensuring that the model expresses appropriate uncertainty in its predictions. The KL term thus balances model complexity against data fit, effectively contributing to the variational objective optimized during training.

The output of these layers is then passed to a final dense layer with a sigmoid activation function, producing a single probability between 0 and 1 that can be used for binary classification.

3.3.2. Probabilistic BNN

The second BNN architecture we develop outputs a probability distribution rather than a point prediction for each input. This allows the model to explicitly represent the uncertainty associated with each prediction. In probabilistic BNNs, predictions for each observation are represented as probability distributions rather than single point estimates. The mean of this distribution is often used as the final predicted probability, while the standard deviation quantifies uncertainty, and a threshold (e.g., 0.5) may be applied to assign a class label. A larger standard deviation indicates lower confidence, while a smaller one indicates higher confidence.

Similarly to our point prediction model, the inputs are concatenated and passed through a batch normalization layer, followed by the same Bayesian hidden layers, priors, posteriors, and KL divergence penalty.

After the Bayesian layers, the resulting hidden representation is passed through a dense layer that produces the parameters for the Independent Bernoulli layer. The Independent Bernoulli layer then defines a full probability distribution over the binary outcomes for each input. Following the Keras tutorial (Keras, 2021), the probabilistic BNN is trained using a negative log-likelihood loss, which directly maximizes the likelihood of the observed data under the predicted distributions.

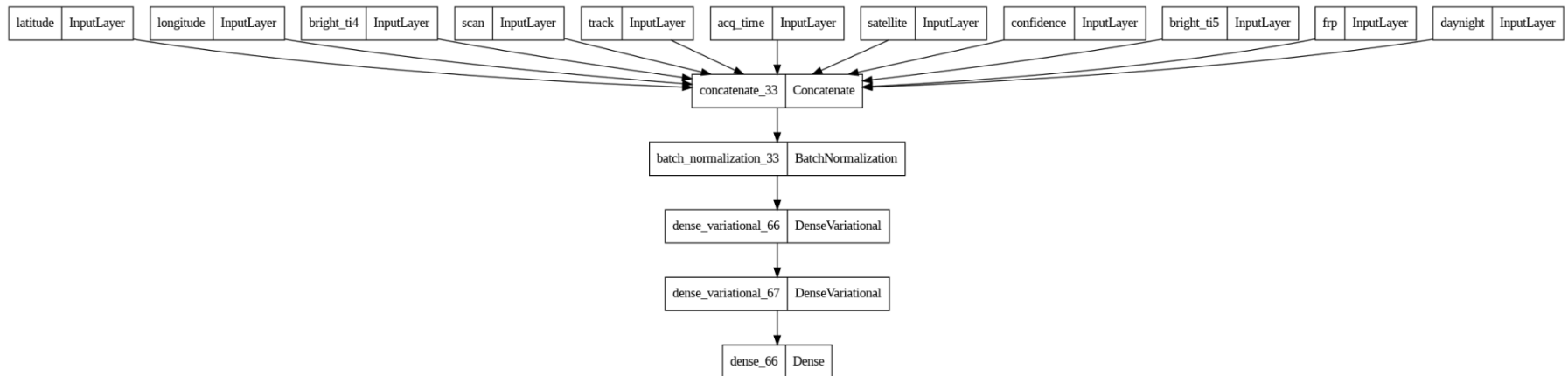


Figure 4. This figure shows the basic BNN model architecture generated by the Keras function plot model. The feature input layer (features are described in Table 1), is followed by a concatenation layer that merges the inputs into a single vector, and then through a batch normalization layer. The normalized output is fed into a series of fully connected (dense) layers. The output is then passed to a final dense layer with a Sigmoid activation function. The probabilistic BNN (Model Two) replaces this layer with an Independent Bernoulli layer. Note, the numerals after the layer name are indicative of how Keras names its model's layers and are not meant to represent a characteristic of that layer type in general.

3.4 Model training

Model development typically involves three distinct datasets: training, validation, and testing. The training and validation datasets are used during the model training process, while the testing dataset remains completely independent and is used only after training to provide an unbiased evaluation of model performance. We follow the direction of the original BNN Keras tutorial, applying an 80/20 split to the original dataset to separate the training and testing data. Within the training set, we further partition the data using Keras' "validation_split" parameter, allocating 80% for training and 20% for validation during each training epoch.

Following the directive of published Keras tutorial (Keras, 2021), a binary cross-entropy loss function is used to evaluate model performance after each epoch for all models, except the probabilistic BNN, which uses a negative log-likelihood loss as the output of the model is a distribution rather than a point estimate. All models are trained using the standard unmodified Adam optimizer as is recommended in the original Keras tutorial (Keras, 2021).

The number of training epochs is typically tuned by monitoring the loss curves and identifying when model performance begins to plateau. This process is applied individually for each case study. As a result, 30 epochs are used for training Models One and Two in Case Study A, while Case Study A Model Three, and Case Study B Models One and Two are trained for 100 epochs.

3.5 Hyperparameter Tuning

We extended the published Keras tutorial (Keras, 2021) by including hyperparameter tuning as a key element. Hyperparameters are model configuration settings that are set before training (as opposed to parameters such as weights, which are learned during training). Tuning these hyperparameters involves modifying their values from their default settings to potentially improve model performance. By selecting more appropriate starting values, the training process may converge faster or result in better final performance compared to relying on defaults. To tune a hyperparameter, multiple models are trained, each using a variation of the target hyperparameter, with the option that produces the highest performing model selected to be used for further training.

Hyperparameter tuning can be done manually, but this approach is often time-consuming and inefficient. A more structured method involves using tools like Scikit-Learn's ParameterGrid (Russell & Norvig, 2022), which generates all possible combinations of hyperparameters from a defined grid. While ParameterGrid itself does not automate model training or evaluation, it simplifies setting up the search space by producing every hyperparameter configuration that must be tested. We then iterate over these combinations, training and evaluating a separate model for each to identify the most effective hyperparameter set. In this study, we use ParameterGrid to explore three hyperparameters: the prior distribution, the activation function, and the number of hidden units. Between three and five values are tested for each hyperparameter, as shown in Table 3.

Table 3: Default and alternative hyperparameters that are tuned within this tutorial.

Parameter	Unique to BNN	Processes influenced	Default	Alternatives	Citation for more information
Prior distribution	Yes	Training	Multivariate NormalDiag	MultivariateNormalDiag scaled by 20, Beta(0,1)	TensorFlow probability distributions
Activation function	No	Training	relu	Sigmoid, gelu, elu, leaky relu	TensorFlow activations
Hidden units dimension	No	Training	8	4, 16	Tensorflow probabilistic layers
Threshold value	No	Evaluation	0.5	0.25, 0.3, 0.35, 0.4, 0.45, 0.55, 0.6, 0.65, 0.7, 0.75 (Optimized value between 0.25 - 0.75 based on maximized performance metrics)	N/A

After training, another important hyperparameter called the threshold value can be adjusted to tailor the model's predictions to specific needs. As previously explained, the model outputs a confidence score representing the probability that an input belongs to class 1. Scores closer to 1 indicate higher confidence in class 1, while scores near 0 suggest class 0. The threshold defines the cutoff point at which the model's confidence is considered sufficient to assign a class label. For example, with a threshold of 0.5, any output score equal to or above 0.5 is classified as class 1, and scores below 0.5 are classified as class 0. While 0.5 is the default threshold, it can be tuned to prioritize metrics such as precision or recall (as discussed in Section 3.6), depending on the application's requirements.

3.6 Model Evaluation

The different models that are generated during hyperparameter tuning are qualitatively evaluated using a Parallel Coordinates Plot (PCP) (Plotly Technologies Inc., 2015), a scatter plot comparing training loss to training validation, and a SHapley Additive exPlanations (SHAP) kernel explainer (Lundberg & Lee, 2017). The hyperparameter combination that produces the model with the lowest validation loss value is then selected to be used to evaluate against the models generated from other runs.

Next, the class label threshold values are optimized for each model. This is done by generating class labels using a range of threshold values with steps of 0.05, from 0.25 - 0.75. Threshold value versus performance metrics are plotted on a multi-line plot, and the threshold value that maximized the most metrics for a model is selected. An accuracy assessment is implemented for every model, producing accuracy, precision, recall, and F1 scores for each set of labeled predictions. Using these optimized threshold values, confusion matrices are generated and finalized accuracy, precision and recall scores are reported. A loss curve is also generated for each model, so the choice of epoch for model training could be qualitatively evaluated.

3.6.1 Hyperparameter plot evaluation

We use a PCP to visualize the effectiveness of hyperparameters. A PCP is a visualization technique that depicts each parameter within a combination as a polyline as it traverses a set of parallel axes (i.e., with the parallel axes in this instance, representing the prior distribution, activation function, hidden units, train loss, and validation loss). As the goal of hyperparameter tuning is to minimize the training and validation loss, the set of hyperparameters with the lowest loss can easily be identified using this technique. Another technique that is implemented in this tutorial is a training versus validation scatter plot. This type of plot is also used to provide visual insights into the best hyperparameter combinations and convergence. The hyperparameter combinations located at the lower-left hand corner of the plot indicate that these models have the lowest validation and training loss. A diagonal, dotted line indicates where training loss equals validation loss and helps indicate convergence amongst the models. Models that are more congruent and are less likely to be experiencing either overfitting or underfitting can be found closest to the diagonal line. When the model is closer to the y-axis and further away from this line, this could indicate overfitting, as the model produces a higher validation loss than training loss.

3.6.2 Accuracy Assessment

Each model's classification accuracy is evaluated using a confusion matrix that compares predicted labels to true labels (Ting, 2017). A confusion matrix contains the assessment results: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) (Olofsson et al., 2014). True Positives are cases where the model correctly predicted a positive class. Conversely, False Positives (FP) refers to instances that the model classifies as positive, but their actual labels are negative (also known as a Type I error). True Negatives (TN) are correctly predicted as negative, while False Negatives are cases in which the model is classified as negative when the actual label is positive (also known as a Type II error). A classification accuracy is evaluated based on how well the mode correctly predicts the labels, this is known as accuracy (Equation 1).

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

Equation 1

A higher accuracy typically indicates a better performing model, but this is not always true, especially in cases of imbalanced classes. Accuracy can be misleading as it gives equal weight to a model's ability to predict all classes. For example, if 90% of labels are negative, a

model that only predicts negative labels, could reach an accuracy of 90%. (i.e.: $\text{Accuracy} = (0 + 90) / (0 + 90 + 0 + 10) = 90\%$)

To strengthen our assessment of the model's performance, metrics such as precision and recall are used. Precision measures how often the model's predictions of the positive class are truly members of the positive class according to the reference datasets (Equation 2). The higher the precision, the better the model becomes at predicting positives. Recall measures how often a model correctly identifies these positive instances from all of the positive instances in the dataset (Equation 3). The F1 score is a harmonic mean of precision and recall. The F1 score tries to balance both the precision and recall and is most often used with imbalanced datasets (Equation 4).

$$\text{Recall} = TP / (TP + FN)$$

Equation 2

$$\text{Precision} = TP / (TP + FP)$$

Equation 3

$$F1\text{ Score} = 2 * ((\text{Recall} * \text{Precision}) / (\text{Recall} + \text{Precision}))$$

Equation 4

3.6.3 Loss and the Learning Curve

Loss is a measure of how far a model's prediction is from the label. The loss function is a mathematical function that helps calculate this. We primarily used the binary cross-entropy loss is used as it is the most commonly applied loss function for binary classification tasks (Equation 5). Binary cross-entropy loss quantifies the difference between the label and the model's predicted probability for that label. It penalizes confident wrong predictions more heavily than wrong predictions that are uncertain, helping to better train the model.

$$BCE(y, \hat{y}) = - [y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$$

Equation 5

where y is the true label and $\hat{y}$ is the predicted label. Binary cross-entropy is not used as the loss function for the probabilistic BNN as binary cross-entropy can not process outputs formatted as distributions, as occurs when using a probabilistic BNN. Instead negative log-likelihood is used as it is a loss function that can be used on probabilistic distributions. Conceptually, negative log-likelihood loss functions work the same as binary-cross entropy by quantifying the difference between labels and the predicted probability.

Models that have a lower loss typically make predictions that better resemble the labels. The loss is updated after each epoch, and by plotting the loss over each epoch (also called a learning curve), we can gain insights into how the model's performance improves over time. When a learning curve is smooth, this indicates the model's performance is consistently and gradually learning, the hyperparameters are well tuned, and the training data is mostly free from noise. Big, sudden jumps could indicate that the model's learning process is unstable and needs to be reviewed. This could be caused by too high of a learning rate, noisy training or validation data, or possible issues with the hyperparameter choices. Though, concurrently, this could also just be due to the loss function. Near decision boundaries, some loss functions can exhibit high variability, especially if the difference between label classifications is minimal.

Typically, to know if a model has trained properly, we look for a plateau in the results which indicates that the model has settled on a loss minimum.

One of the main goals of model training is to train that model to perform well not only on the data it is trained on, but also on new data. If a model is able to perform well on training data, but does not generalize well to new data, this is referred to as overfitting. Overfitting can be seen in loss curves when the validation loss starts to diverge away from the training loss. When this occurs, the model typically finds dependencies in the training dataset that are not applicable to new data. This results in high variance, which means the model is highly sensitive to small changes in the dataset. The opposite of this, is referred to as underfitting, when the model performs poorly on training data and generalizes poorly. This typically occurs when the model is not complex enough or the model is not sufficiently trained. This results in high bias, which means the model created incorrect assumptions during learning.

3.6.4 Hypothesis Review

Case Study B evaluates its hypothesis using the standard deviation of the probabilistic BNN's output probability distribution as a proxy for predictive uncertainty. Standard deviation quantifies the spread of values in a distribution: higher values indicate greater spread from the mean, while lower values indicate that values are tightly clustered.

The null hypothesis being tested is that if a model is trained on a multi-source dataset, then its predictive uncertainty should remain consistent across data sources, even when predicting on previously unseen samples. To evaluate this, predictions from Case Study B's Model Two are grouped based on the source of each observation (MODIS or VIIRS, respectively). For each prediction, the standard deviation of the output probability distribution is calculated. A two-sample Student's t-test is then performed to compare the distributions of these standard deviations between the two source groups, assessing whether there is a statistically significant difference in model uncertainty across data sources.

3.5.5 Explainable Artificial Intelligence

Traditionally, it has been difficult to understand how machine learning models conduct their decision making and prediction process. Recent developments, along with a push for understanding these processes, has led to eXplainable artificial intelligence (XAI). This tutorial incorporates SHAP, which is an approach to explain the outputs of machine learning models by leveraging game theory. Fundamentally, SHAP helps measure the contributions of each input to the final outcome. The x-axis indicates the value for a particular data point of a feature, with its color encoding its raw value. A positive SHAP value for a data point suggests that feature helps push the model towards a "fire present" prediction, with a "higher, more positive value" indicating that it influences the model to have a stronger, positive likelihood. While a negative value pushes it towards "no fire".

3.7 Case Studies

3.7.1. Case Study A: Pre-screening VIIRS Active Fire data using a Bayesian Neural Network (BNN).

The goal of the supervised Bayesian neural network from Case Study A is to predict whether a satellite-detected hotspot corresponds to a confirmed fire on the ground. This requires a training dataset containing both verified fire detections (true positives) and

misidentified detections (false positives). We use a systematic Python method to label hotspots as true or false.

Once the hotspots within Canada are obtained, the NBAC polygon, CNFDB points, and persistent hotspot shapefiles are loaded and subset to ensure only data from one day in 2023 are included in the analysis. These shapefiles are then buffered by a predefined distance to account for satellite resolution. For this study, a buffer radius of 750 m, twice the nominal resolution of the VIIRS active fire product, is applied. This threshold is selected arbitrarily as a first pass, and more rigorous buffering approaches are discussed in Section 5.

A separate spatial join is used with the buffered perimeters and hotspots within Canada to determine which hotspots fell within known fire perimeters. Hotspots within buffered perimeters are labelled as true positives (1) and those outside of buffered perimeters are labelled as false positives (0). Figure 5 displays an example of the true positive and false positive classifications for a buffer distance of 750 m. FIRMS offers latitude, longitude, brightness temperature Ti4, brightness temperature Ti5, scan, track, acquisition time and date, satellite name, version, confidence score, FRP, and day/night values for each reported hotspot (features as described in Table 1). Categorical columns are dropped, excluding “daynight”, “satellite”, and “confidence” which are encoded to numerical values. The resulting data is then exported as a CSV for use in this tutorial.

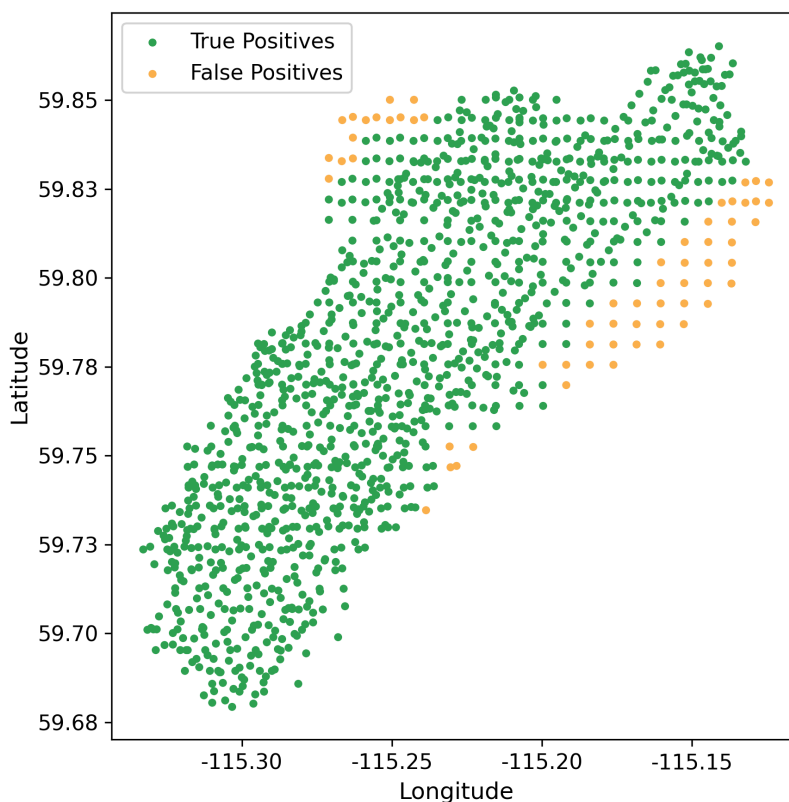


Figure 5. An example of true and false positive classifications for NBAC ID 2123 (Projection: WGS84). True positives, displayed in green, are defined as those hotspots within the buffered perimeter, and false positives, displayed in orange, are defined as those hotspots outside of the buffered perimeter.

Model One

Model One closely follows the original Keras tutorial (Keras, 2021) and does not include any hyperparameter tuning. It uses minimally processed or raw, input data, defined here as data obtained directly from FIRMS, with no preprocessing beyond the basic steps outlined in Section 2.2. The purpose of this model is to serve as a baseline, demonstrating model performance without additional optimization or customization.

Model Two

For the second model, we expand on the tutorial code by including more advanced data preprocessing functions. When applying machine learning methods, it is crucial to understand the characteristics of the input data, including any inherent limitations and biases. In Model One, no effort is made to balance the classes, which can significantly impair the network's ability to learn the key features needed to distinguish between true and false positives (Lee et al., 2016). In Model Two, we address this issue by balancing the dataset before training the model. To achieve this, we apply random undersampling to the original dataset, ensuring balance not only between classes (true and false positives) but also between satellite sources (NOAA-20 and Suomi-NPP) (Figure 6). This helps prevent the model from learning satellite-specific noise patterns or resolution/angle biases instead of features associated with actual fire detections. Once data preprocessing is complete, model compiling and training are implemented in the same way as the previous model.

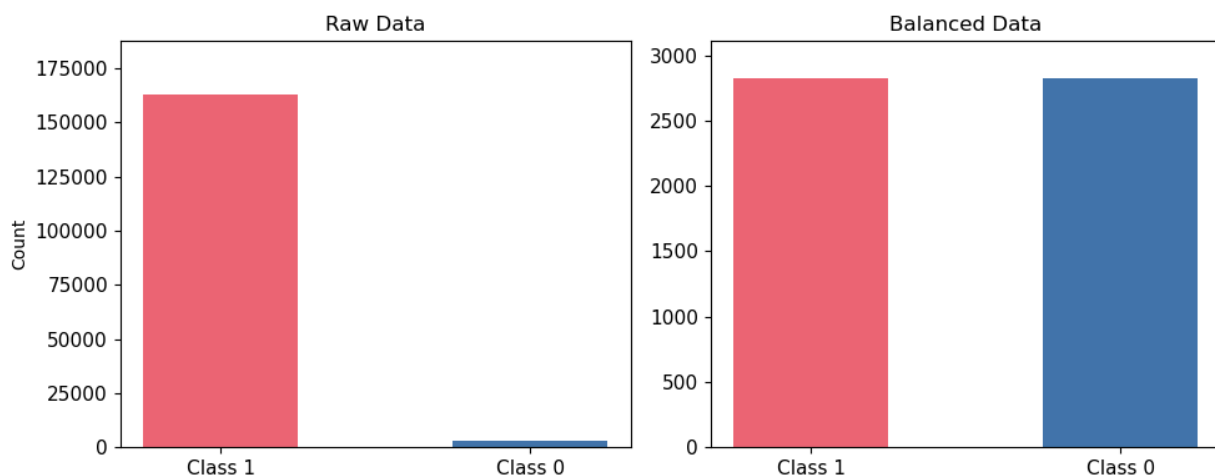


Figure 6. This figure indicates the raw class distribution (used in Model One) versus balanced class distribution (used in Model Two). Class 0 represents false positives and Class 1 represents true positives.

Model Three

For the third model, we use the balanced data from Model Two, and expand the model-building code beyond what is provided by the tutorial to include hyperparameter tuning into the workflow. Three hyperparameters related to model building are identified (Table 3), resulting in 30 models being trained to represent every combination of hyperparameters available. The model that performs best out of this group is then used to compare this run to the

models generated from the previous runs. We also explore the optimization of an additional hyperparameter, the threshold value, used for model evaluation (Table 3). Due to its location downstream in the workflow from model training, adjusting this hyperparameter does not require us to repeat the training portion of the code, only the evaluations section.

3.7.2. Case Study B: Assigning multi-source Earth observation Active Fire confidence using Bayesian Neural Network.

The goal of Case Study B is to examine the impact observation source has on model output as well as demonstrate the impact of upgrading the model to be a probabilistic BNN. The null hypothesis for this case study is that the uncertainty in the classification of an observed hotspot does not differ significantly in regard to the observation source. Case Study B analyzes fire hotspots detected by the VIIRS 375 m active fire product from the Suomi NPP and NOAA-20 satellites, alongside hotspots from the MODIS active fire product onboard the Aqua and Terra satellites. The data preparation methods are consistent with that of Case Study A, with the primary difference being the inclusion of standard-quality MODIS hotspots in addition to those from VIIRS. While the dataset is balanced between MODIS and VIIRS detections, it is not balanced by individual satellites (SNPP, NOAA-20, Terra, Aqua) in order to retain the maximum amount of data. For the purposes of Case Study B, as many of the included variables within the FIRMS datasets are not directly comparable between the MODIS and VIIRS instruments (discussed further in the Discussion section), we retain only FRP, latitude, longitude, scan, track, and daynight values and assume these are comparable. We also retain confidence values by mapping VIIRS categorical confidence levels to approximate percentage values: low to 33%, nominal to 66%, and high to 100%.

Model One

Model One uses the same architecture and hyperparameter tuning steps as Case Study A Model Three, but is trained with the updated data to contain both VIIRS and MODIS. This model is not sophisticated enough to be evaluated for prediction uncertainties but acts as a context for how a non-probabilistic BNN can perform on the new dataset.

Model Two

Model Two introduces a new architecture, a probabilistic Bayesian neural network (BNN), which outputs predictive distributions rather than single point estimates. From these distributions, we extract the mean as the predicted value and the standard deviation as a measure of model uncertainty. These predictions are grouped based on the source of the original observation (MODIS or VIIRS). A Student's t-test is performed to compare the mean standard deviation between groups. A statistically significant result would suggest that the model's uncertainty differs depending on the data source. To further explore this, we generate histograms of the standard deviations for each group to visualize their distributions. Table 4 summarizes all case studies and models.

Table 4. Summary of case studies and models.

Case Study	Model #	Dataset	Point or Probabilistic Prediction	Hyperparameter Tuning
A (VIIRS only)	1	Unbalanced	Point	No
	2	Balanced		No
	3	Balanced		Yes
B (VIIRS and MODIS)	1	Balanced	Point	Yes
	2	Balanced	Probabilistic	No

4. Results

4.1 Case Study A

This section reviews the threshold, accuracy assessment, loss and learning curve, hyperparameter tuning and SHAP values for Case Study A.

4.1.1 Hyperparameter Plot Evaluation

A key aspect of a model's performance is the selected hyperparameters. In our BNN, we focus on tuning the prior distribution function, activation function, and the number of hidden units. The importance these hyperparameters play can be seen in Figure 7 and 8 below. From referencing Figure 8, we find the hyperparameters that create the lowest training and validation loss. From the images, we can see that two models seem to produce the lowest training and validation loss. The first model has a fixed, zero-mean unit-variance multivariate normal prior distribution, activation function of "sigmoid," and hidden units dimension of 4 and the second model has a fixed, zero-mean unit-variance multivariate normal prior distribution, activation function of "relu," and hidden units dimension of 8. If you review the log outputs, the first model has the hyperparameters that result in the lowest train and validation loss.

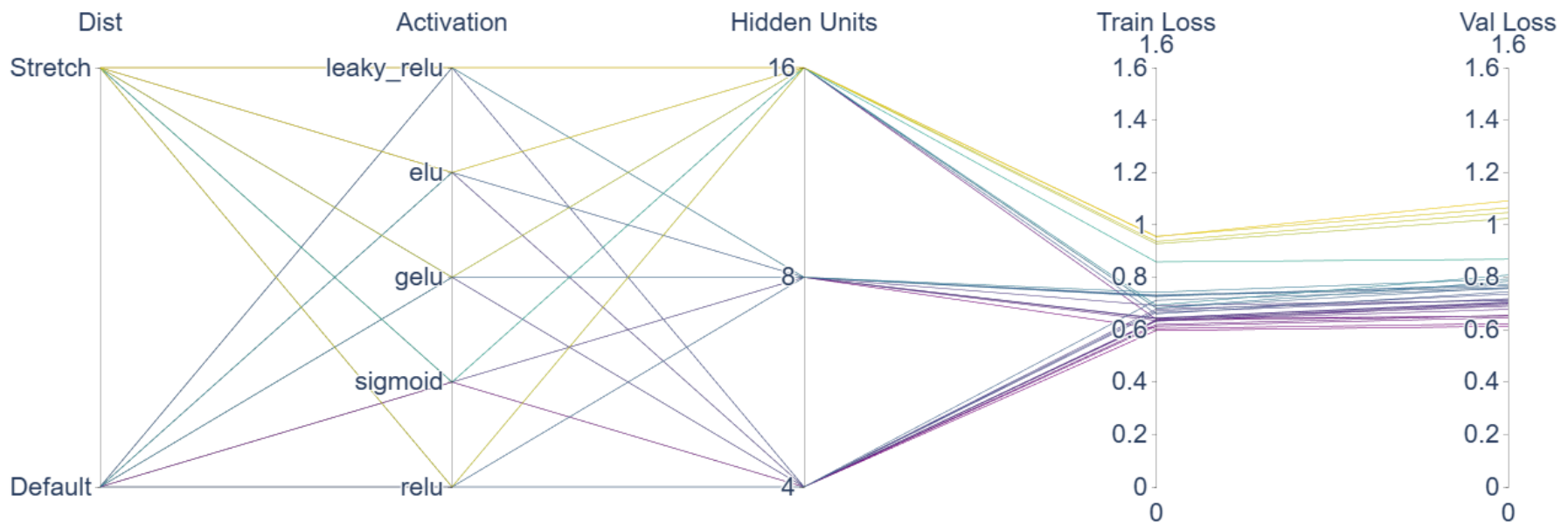


Figure 7: Visual example of a Parallel Coordinate Plot where each line represents a specific combination of input variables (e.g., hyperparameters) that were tested. This plot allows viewers to track how different combinations of input variables perform in relation to each other from both a resulting training and validation loss perspective.

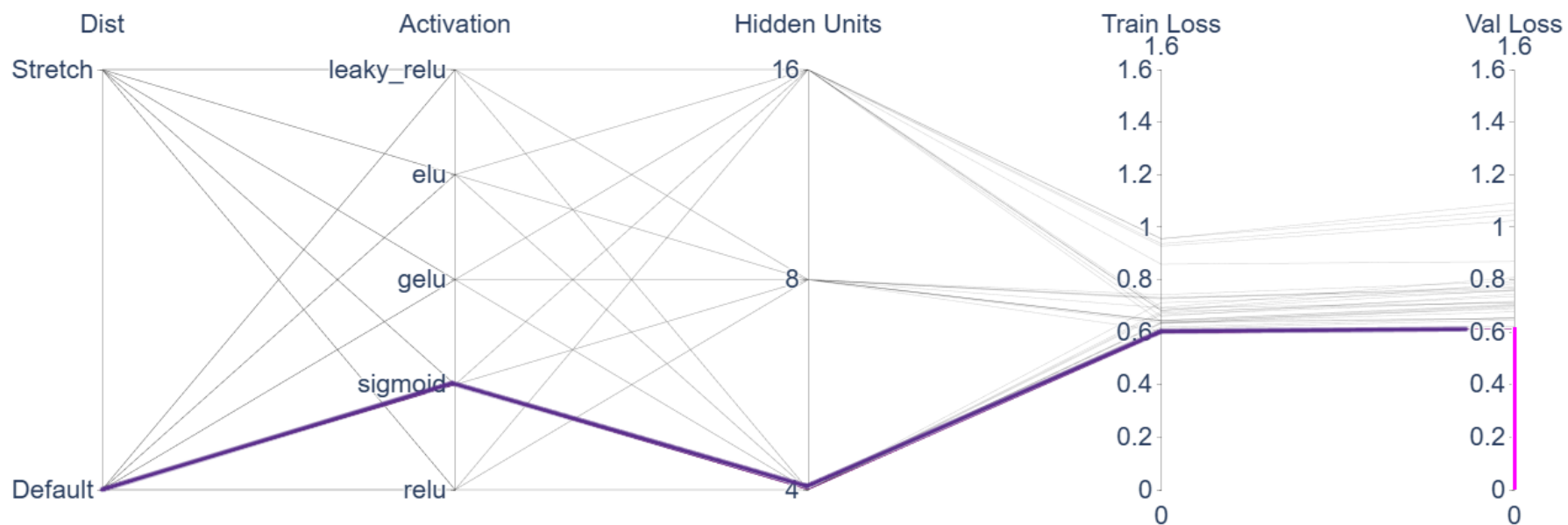


Figure 8: In the Colab notebook, you can interactively select any of the lines (polyline) across the Parallel Coordinate Plot to highlight only it. Tracing this polyline backward across the axes reveals the specific hyperparameter combination associated with the best-performing model, providing insight into which settings minimize both training and validation loss.

As indicated in the training versus validation scatter plot (Figure 9), some of our models experience slight overfitting, specifically the “Leaky relu”, “relu”, “gelu”, and “ELU” activation functions in combination with the prior distribution of “stretch.” Those in the bottom left hand corner have the lowest training and loss validation. The model with the prior distribution of “default,” activation function of “sigmoid,” and hidden units dimension of 4 has the lowest train and validation loss in this plot as well.

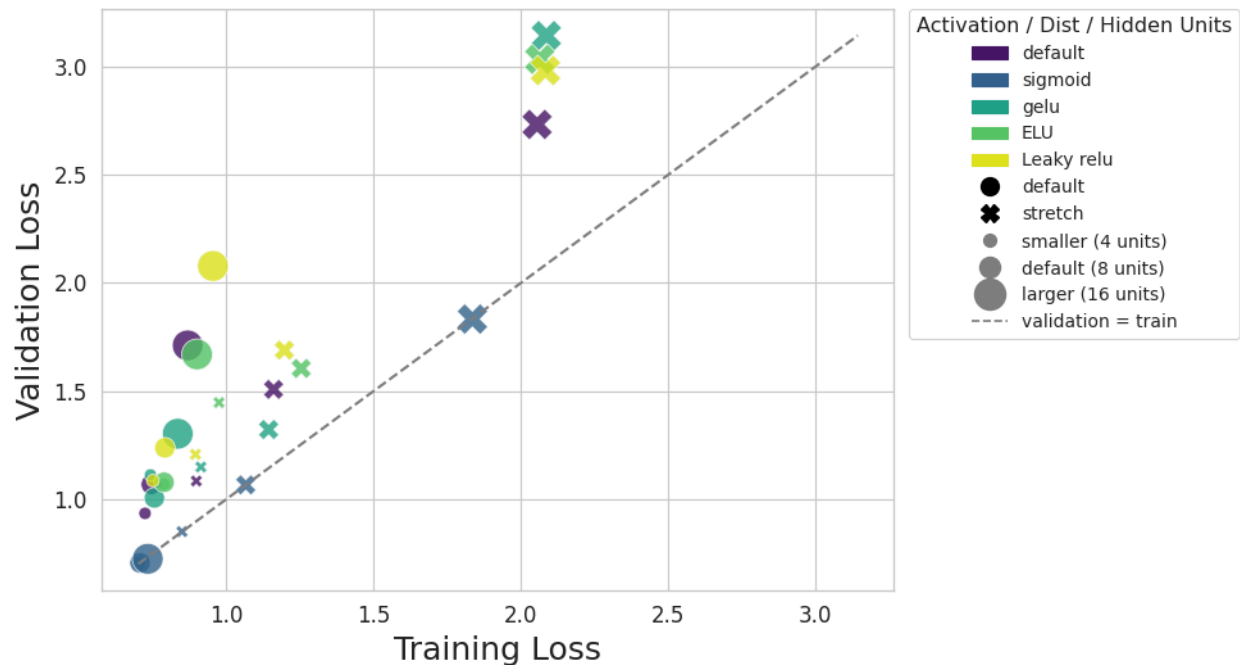


Figure 9: Visualization of the train and validation loss.

4.1.2 Threshold Review

Precision and recall curves can help to indicate what thresholds result in the optimal model performance. Reviewing Figure 10, Model One has maintained high accuracy, precision, recall and f1 values across all thresholds. Model Two experiences an increase in precision and accuracy at 0.48 but a decrease in recall and F-1 at 0.5. Model Three experiences an overall decrease in recall from 0.1, an increase in accuracy until 0.55 and an increase in F1 until 0.45 where they both then plateau, and a general increase in precision throughout all thresholds.

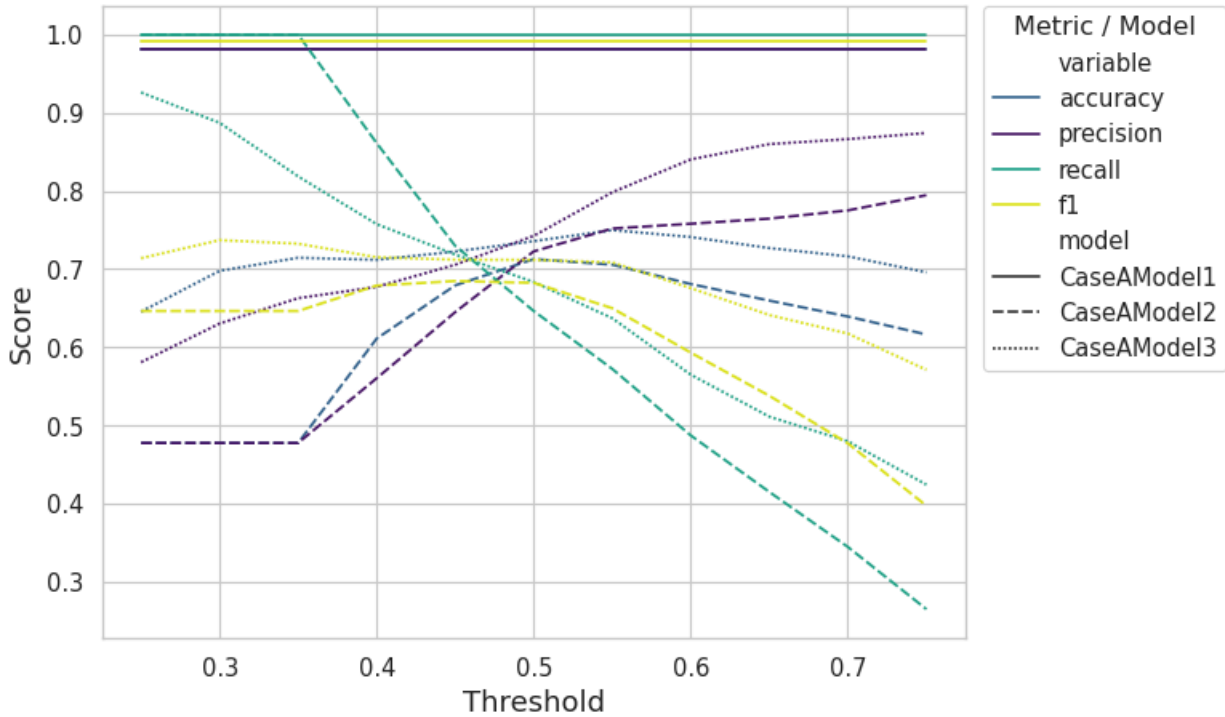


Figure 10: We graphed accuracy, precision, recall, and f1 values across all three models for Case Study A.

4.1.3 Accuracy Assessment

We used the values identified above as the thresholds to compute a confusion matrix for each model (Figure 11). Model One remains unaffected by threshold so we set its threshold to a default value of 0.5. Case Study A Model One achieves an overall accuracy of 0.98, with a precision of 0.98 and a recall of 1.0. It never predicts label 0, it only predicts label 1 values; and delivers 98.21% correct predictions with 1.79% as false positives. Model Two achieves an accuracy of 0.71, a precision of 0.72 and a recall of 0.65. It heavily predicted more label 0 than label 1, with a label 0 prediction of 57.30% (40.41% are correct predictions and 16.89% are not). Of the label 1 predictions, 30.86% are correct predictions while 11.85% are false positives. Model Three has an accuracy of 0.73, a precision of 0.70 and recall of 0.78. The model predicts label 1 (53.93 %) slightly more than label 0 (46.07%), with 37.40% as true positives and 16.53% as false positives.

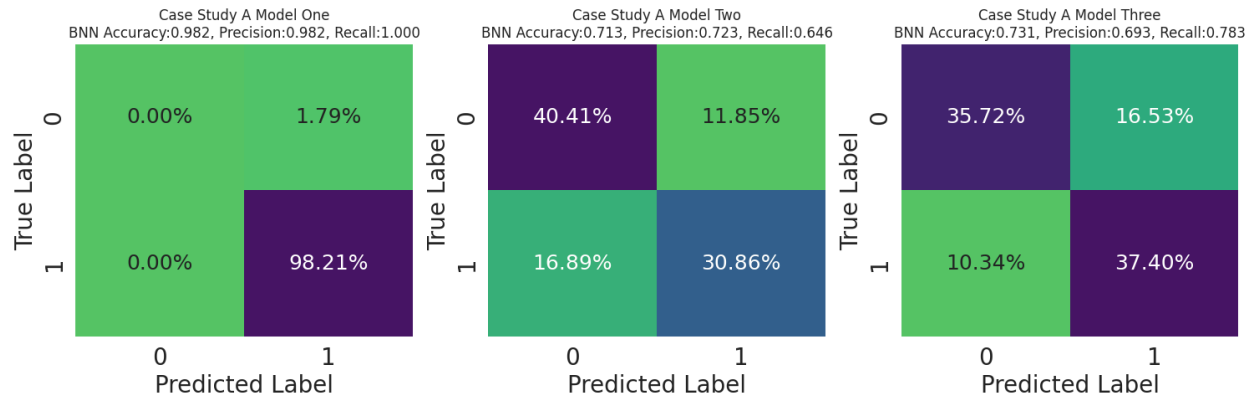


Figure 11: This confusion matrix contains the accuracy, precision and recall for all three models of Case Study A.

4.1.4 Loss and the Learning Curve

Figure 12 displays three plots that illustrate the training and validation loss curves over epochs for Model One, Model Two, and Model Three within Case Study A. In all three cases, the training and validation loss decrease relatively rapidly in the initial epochs, indicating that the models are learning effectively. For Model One and Model Two, the training and validation loss curves are similar throughout the training process, suggesting that these models are generalizing well to unseen data and are not overfitting. If we combine this with the knowledge that the model is predicting a single class, it means the model is converging to a trivial solution where it minimizes loss by guessing the most frequent class, rather than understanding the underlying data. In Model Three, while both losses initially decrease, the training loss eventually continues to decline, whereas the validation loss begins to plateau. This divergence, where training loss continues to improve but validation loss stagnates or worsens, is a clear indication of overfitting in Model Three, as the model is starting to memorize the training data rather than learning generalizable patterns.

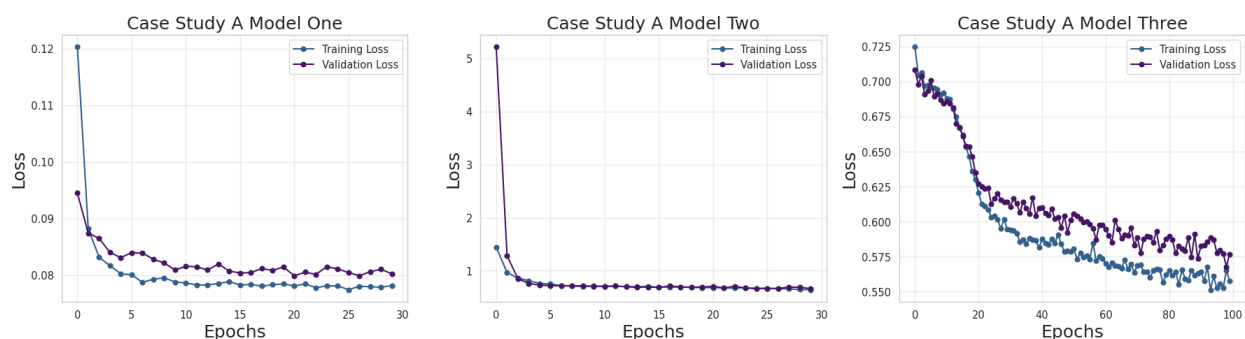


Figure 12: This figure shows the training and validation loss curves for all three models in Case Study A.

When reviewing the loss curve, it is essential that we do so in tandem with the confusion matrix. While a plateauing of the loss curve typically indicates convergence, sometimes this can be misleading as it does not always indicate a well rounded model. Model One is a great

example of this. Model One where the loss curve indicates a well fitted model, but it completely ignores label 0 (0% predicted) and only focuses on predicting label 1 (98.21% true positives predicted). This could indicate that the model is updating the loss function by predicting label 1, which would (falsely) minimize the loss. This falsely minimized loss would lead to a loss curve that would mimic one of a well fitted model, causing confusion for the modeller at first glance. Loss curves are used as a diagnostic tool in tandem with other methods to ensure a strong model.

4.1.5 Explainable Artificial Intelligence

Figure 13 contains the SHAP for Model Three of Case Study A and it visualizes which of our input values are of most importance to the model. The “bright_ti5” (Table 1) input has the biggest spread across the x-axis. Additionally, the data points with the larger values influence the model toward predicting “fire,” while data points with smaller values influence the model toward predicting that the pixel does not contain fire. The “satellite” and “confidence” inputs have most of the data points clustered around the zero value, indicating that the detection algorithm confidence does not have a particular impact on the model.

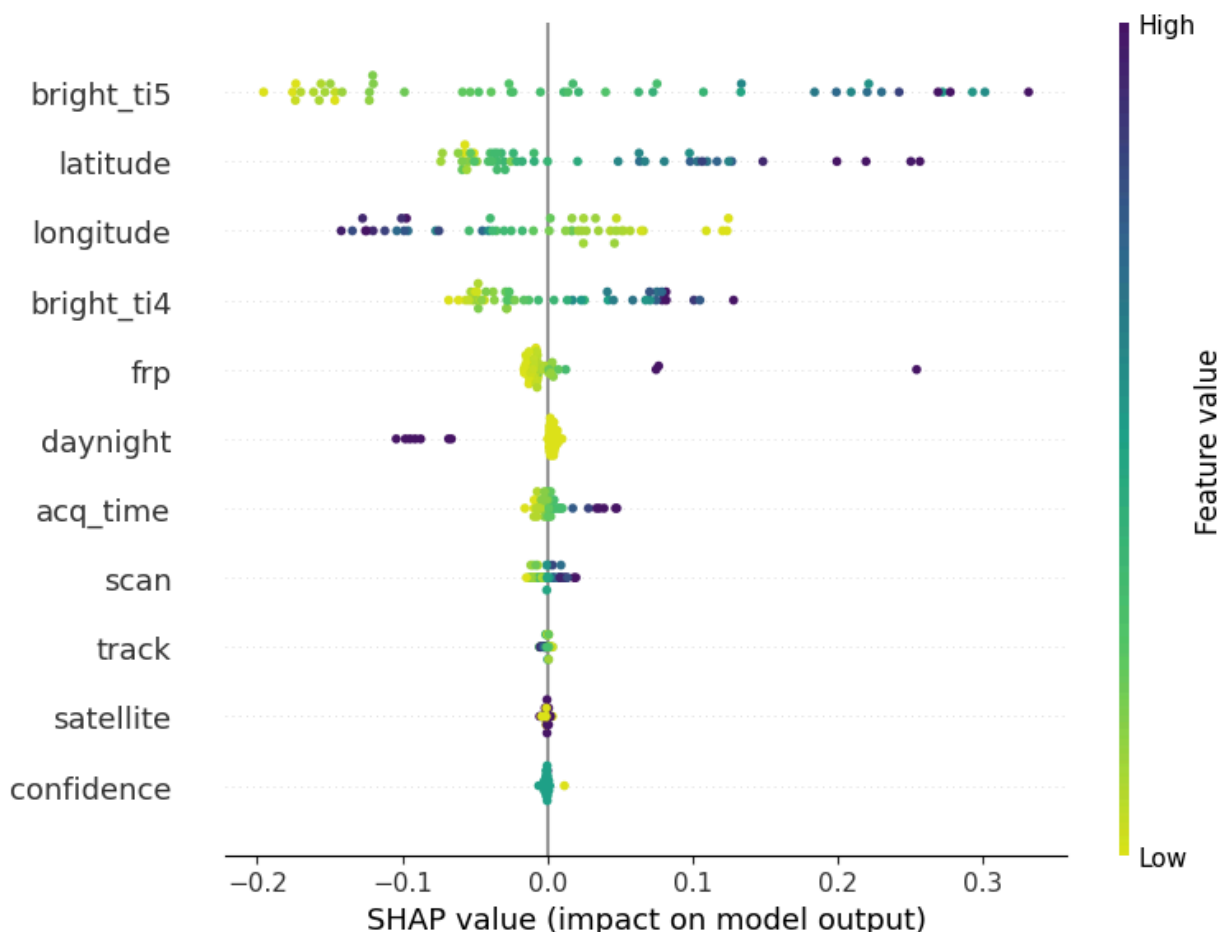


Figure 13: This SHAP plot shows how each input influences the model predictions for Case Study A Model Three.

4.2 Case Study B

This section reviews the threshold, accuracy assessment, loss and learning curve, hypothesis review, and SHAP values for Case Study 2.

4.2.1 Threshold Review

Figure 14 displays Case Study B, Model One initially achieves its highest accuracy at the threshold of 0.55 and then levels off, while seeing a significant drop in precision after the 0.5 threshold. At roughly 0.45 and 0.4, it experiences a steep decline in recall and F1 respectively. However Model Two stays constant across all thresholds. We select a threshold of 0.5 for both Model One and Model Two.

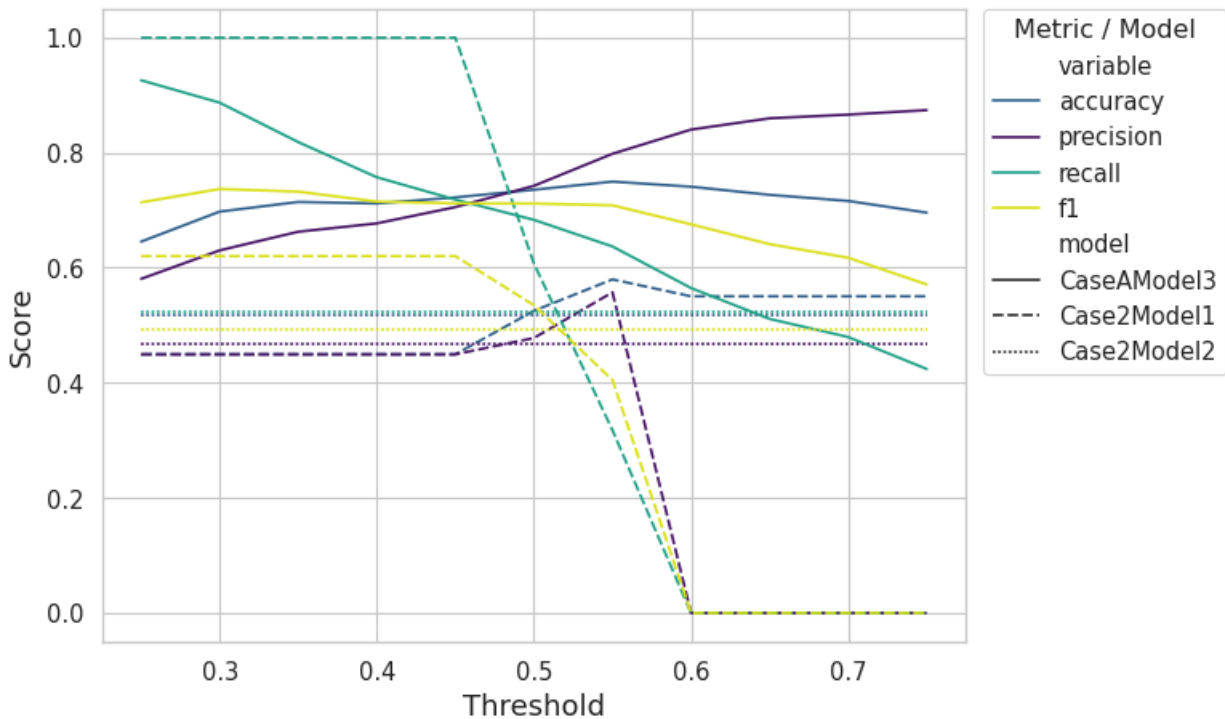


Figure 14: This figure shows the accuracy, precision, recall, and f1 values for both models for Case Study B. Case Study A Model Three is included for reference.

4.2.2 Accuracy Assessment

The results of Case Study B using Model One show an overall accuracy of 0.53, a precision of 0.44 and a recall of 0.22 (Figure 15). From the confusion matrix heatmap in this figure, we can see that the model correctly predicts label 0, 42.44% are true negatives, while 12.61% are false positives. It heavily predicted more label 0 than label 1, with a label 0 prediction of 77.31% (42.44% are correct predictions and 34.87% are not) and struggles to predict label 1 at 22.69%. Of the label 1 predictions, 10.08% are correct predictions while 12.61% are false positives. An overall accuracy of 52.5% indicates that this model might not be very effective.

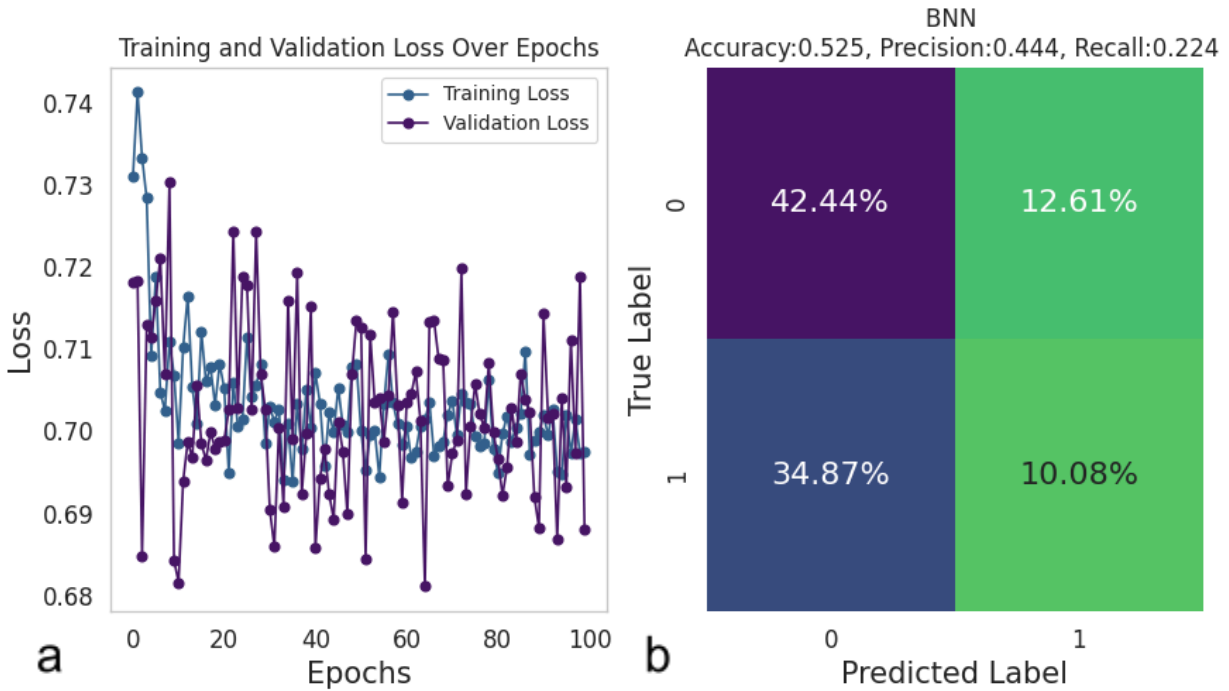


Figure 15: Accuracy assessment of Case Study B, Model One showing a) training and validation loss curve over epochs, and b) confusion matrix.

The results of Case Study B with Model Two include an overall accuracy of 0.55, a precision of 0.49 and a recall of 0.71 (Figure 16). From the confusion matrix heatmap in Figure 15, we can see that the model correctly predicts label 0 only 22.69% of the time while incorrectly labeling them 32.35% of the time. For label 1, the model is correctly predicting 31.93% of the time and incorrectly 13.03%. The model tends to predict label 1 (64.28% overall; 31.93% correctly, 32.25% incorrectly) less often than label 0 (35.72% overall, 22.69% correctly, 32.35% incorrectly). Though the overall accuracy is relatively poor (54.6%), with a recall of 71%, the model seems to be accurately capturing the majority of positives but is predicting a significant amount of false positives (low precision of 49.7%).

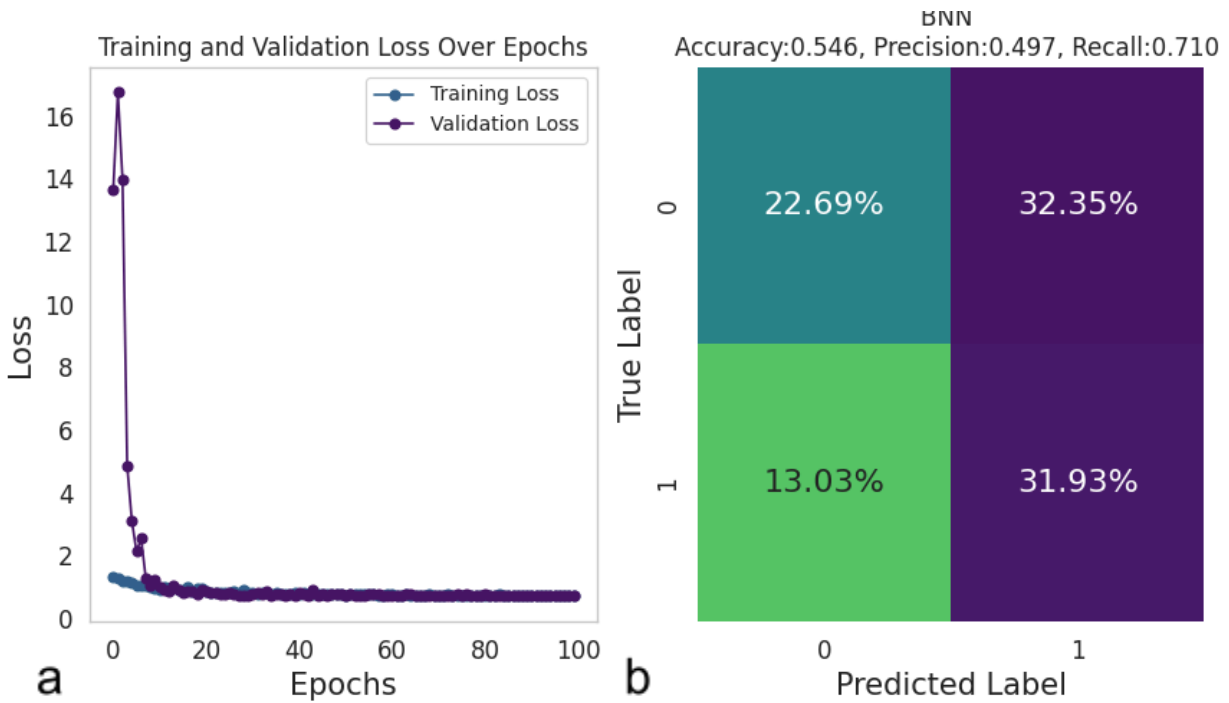


Figure 16: Accuracy assessment of Case Study B, Model One showing a) training and validation loss curve over epochs, and b) confusion matrix.

4.2.3 Loss and the Learning Curve

The loss curve for Case Study B with Model One (Figure 15) shows a general decrease over the epochs, with no clear signs of overfitting nor underfitting. The training and validation loss curves show light fluctuations mostly within a range of 0.68 to 0.72, which are typical in neural network training and may reflect minor noise in the data, sensitivity to the learning rate, or inherent variability in the model's optimization process. Though, after 100 epochs, this is still relatively high and might indicate it is not done training. Generally, the loss curve for Case Study B with Model Two in Figure 16 seems to indicate strong improvement training and validation loss. When reviewing the loss curve, validation loss is initially extremely high (slightly over 16), but drops quickly within the first five epochs and fluctuates around 1, indicating that it most likely predicts incorrectly in the beginning but then learns to correct itself. After epoch 5, training and validation seem to converge and remain consistent indicating that they have either reached an optimization plateau (i.e.: they are unable to find a weight distribution to improve) or have reached the most optimal solution they can find based on the data.

4.2.4 Hypothesis Review

For Case Study B with Model Two, we also review the p-value from the Student's t-test to understand if the source impacts the prediction confidence. The p-value is $3.1e^{-5}$ and since it is below 0.05, we can reject the null hypothesis and accept that the source does impact the standard deviation of the prediction confidence.

As discussed in Section 3.4.4, for each input, the probabilistic BNN will output a distribution of probabilities rather than a single point estimate. For each output distribution, we calculate the mean value as the final predicted probability and assign a threshold above which

to define the classification label, and also calculate the standard deviation of each output. These are plotted in Figure 17. The BNN appears to have learned a very specific and consistent prediction for "source 1", with most of its predictions for this source falling within a range of 0.10. For "source 0", the model's predictions are more diverse, indicating that the data points from "source 0" might be inherently more varied or that the model is less certain about a single "typical" prediction value for this source. If a classification threshold (e.g., 0.5) were applied, a large majority of "source 1" would fall on one side of the threshold, whereas "source 0" would have a more even split or be more distributed across the threshold, leading to more ambiguous classifications for "source 0".

Both sources generally exhibit high uncertainty from the BNN, especially at the 0.50 mark. This high standard deviation suggests that for many predictions, the model's posterior distribution over the output probability is broad, implying that it's not very confident in a single precise probability value. However, the key difference lies in the presence of lower standard deviations for "source 1". This indicates that while the BNN is often highly uncertain for both sources, there are cases where it feels more certain about "source 1" predictions than it ever does for "source 0". "Source 0" appears to consistently elicit maximum uncertainty from the model, whereas "source 1" occasionally allows the model to have instances of higher confidence (lower standard deviation).

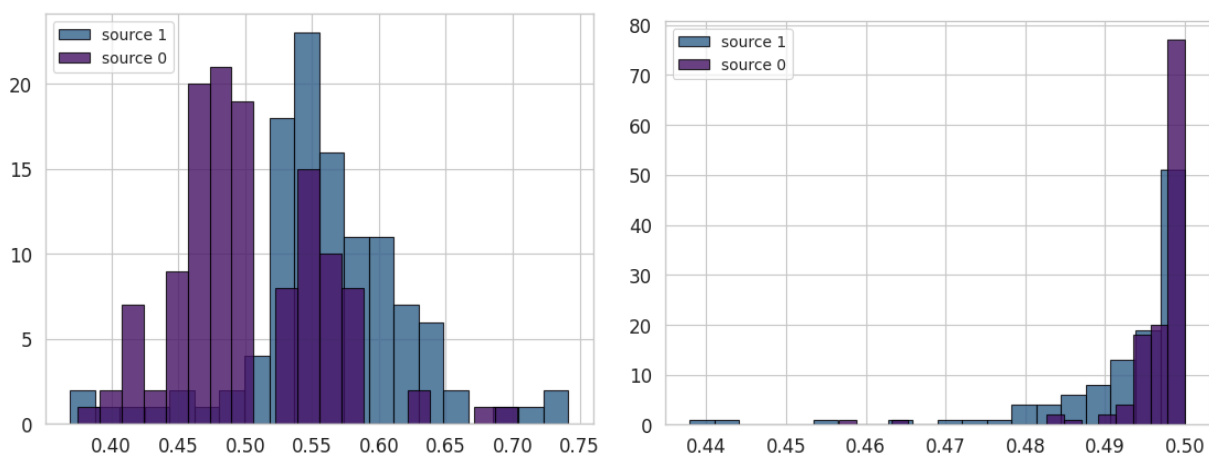


Figure 17: Prediction means a) and standard deviation b) of Source 0 (VIIRS) and 1 (MODIS) for Case Study B using Model Two.

4.2.5 Explainable Artificial Intelligence

In order to understand exactly what the model is learning, a SHAP is also calculated for Case Study B with Model Two (Figure 18). Unlike in Case Study A with Model Three, the SHAP values are more tightly clustered around 0.0, with the min and max being between -0.10 and 0.10. This demonstrates that the features do not influence the model as strongly, as they do in Case Study A with Model Three. This could denote that the model is not as confident and is not very predictive from this data. Though the "source," "scan," and "latitude" do have some slightly higher values that seem to impact the model.

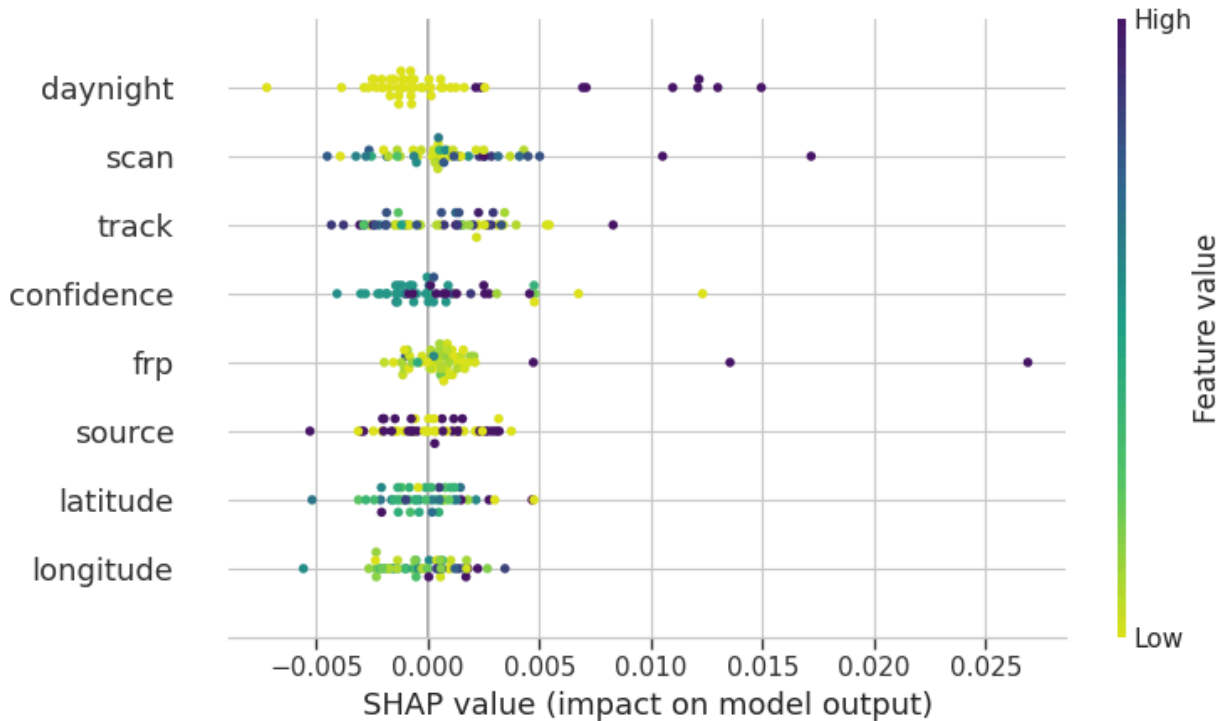


Figure 18: SHAP features and their impact on the model predictions for Case Study B Model Two. See Table 2 for definitions of features.

5. Discussion

This study explores the application of deep learning techniques, particularly BNNs, to model wildfire activity, highlighting key insights into model interpretability, uncertainty quantification, and the practical challenges of real-world implementation. Our analysis underscores that while traditional accuracy metrics can be misleading, tools like hyperparameter tuning and explainability plots (e.g. SHAP and parallel coordinates) are crucial for building robust and trustworthy models. Bayesian networks not only provide predictions, but also can provide crucial measures of reliability and spread, offering a more comprehensive understanding of model uncertainty, which is vital when comparing results across diverse data sources.

5.1 Models and Results

Our exploration within this notebook reveals several critical takeaways for deep learning users. Firstly, the significant positive impact of hyperparameter tuning on model performance is evident. Model One, trained on an imbalanced dataset and without hyperparameter tuning, achieved an artificially high accuracy by exclusively predicting class 0, rendering its results misleading and ultimately untrustworthy. In contrast, Model Two, while also not undergoing hyperparameter tuning, was trained on a balanced dataset, producing more reasonable and trustworthy results than Model One, serving as a valuable baseline. Model Three, however, demonstrated the true power of optimization: trained on a balanced dataset and leveraging hyperparameter tuning, it achieved superior performance with higher accuracy and recall compared to Model Two, reflecting the most robust and reliable understanding of the data.

Secondly, we demonstrate that accuracy alone can be a highly misleading metric, especially in the context of imbalanced datasets, as illustrated by Model One. This underscores the necessity of employing more robust and comprehensive evaluation strategies.

Finally, probabilistic Bayesian Neural Networks (BNNs) offer a superior approach to understanding prediction reliability and uncertainty compared to standard neural networks. A standard neural network typically produces a single point prediction without quantifying the inherent uncertainty. Probabilistic BNNs, by placing probability distributions over their weights and biases, become probabilistic models that provide a distribution of possible predictions, from which we can derive confidence estimates and intervals. This allows for a standardized measure of uncertainty, facilitating more meaningful comparisons across different data sources and enhancing the explainability of results to non-technical stakeholders, ultimately fostering greater trust in decision-making processes.

However, the practical implementation of these sophisticated models within a tutorial setting also exposes inherent limitations, largely driven by computational constraints. The training of our models is significantly curtailed by the limitations of Google Colab, specifically concerning training duration, permissible data volume, and model complexity. The loss curves, which do not plateau in some cases, strongly suggest insufficient training, indicating that the models had not fully converged. With more computation power, we would have been more able to experiment with implementing other optimization methods such as early stopping, where training halts if the validation loss does not improve for a predefined number of consecutive epochs (patience). This helps prevent overfitting by stopping training once the model stops making meaningful gains. Additionally, we train on only a single day's worth of data; which still consumes a substantial amount of computational resources. This compromise was done to demonstrate the feasibility of the models and we do not intend it to be used for practical deployment, as it restricts the model's ability to generalize across temporal variations. For complex machine learning and DL techniques, especially when dealing with large datasets or intricate architectures, access to supplementary computational resources is essential.

5.2 Limitations and Impact

If this methodology were to be extended to a real-world study, several key adjustments and enhancements would be required to improve the robustness, accuracy, and practical relevance of the models. These improvements can be broadly grouped into three areas: data preprocessing refinements, expanded data coverage, and incorporation of domain expertise. Several real-world applications of Bayesian Neural Networks (BNNs) already exist outside the wildfire domain. For instance, Ma et al. (2021) used BNNs to predict corn crop yields in the United States, while Nguyen et al. (2023) employed BNN-LSTMs (Bayesian Long Short-Term Memory networks) to forecast air pollution in Australia. Readers are encouraged to consult these studies for further insight into practical implementations of BNNs.

5.2.1 Data Preprocessing

Our current approach involves several simplifying assumptions. The assumption that any hotspot outside the final fire perimeter is a false positive needs refinement. We would, for instance, mislabel smaller fires, such as intentionally set agricultural burns that are not captured in datasets like NBAC. A more sophisticated approach would involve flexible buffer distances informed by FIRMS scan and track values, accounting for satellite resolution variations (e.g., buffering by the diagonal length of the pixel) to ensure more accurate perimeter-hotspot associations. Furthermore, while random undersampling helped mitigate class bias, it led to a substantial reduction in sample size, potentially limiting the model's ability to capture subtle features and increasing the risk of overfitting. Future work would prioritize alternative balancing techniques, such as synthetic oversampling, to preserve data integrity and enhance

generalizability. For Case Study B, although deep learning can accommodate some variability between data sources, particularly through standardization and normalization, the substantial differences between MODIS and VIIRS active fire products (e.g., spectral bands for brightness temperature, confidence score representation, spatial resolution, and FRP calculation methodologies) still necessitate careful harmonization to ensure that the model learns meaningful patterns rather than artifacts of the data sources.

Our current data processing methods use an internal persistent hotspots shapefile, but future users may use a similar dataset containing global static thermal anomalies which was made available in NASA FIRMS in March 2025 after extensive engagement with international fire scientists and managers (D. Davies, 2025).

5.2.2 Data Coverage and Quality

The current model, which we train on data from only one day of a single year that is specific to Canada, has limited generalizability. An operational quality study would require an expanded temporal scope to multiple years and multi-date time series data, providing a richer training dataset and enabling the model to learn more robust patterns across time. Collaborating with local experts would be invaluable for expanding spatial coverage to other fire-prone regions and, crucially, for refining label quality. The observed sensitivity of the model to thresholding highlights the importance of accurate and unbiased labels. Leveraging expert knowledge in creating thresholds and validating labels would significantly enhance the trustworthiness of the training data and, consequently, the model's results. Additionally, incorporating a broader range of supplementary data sources, such as meteorological data (e.g., Meteosat Third Generation), detailed land use/land cover information, and infrastructure data, could further enrich the model's understanding of environmental conditions that influence true and false fire positives. Finally, as MODIS approaches its end-of-life, supplementing it with newer data sources and integrating additional user and ground validation would ensure the continuous relevance and accuracy of the data used.

5.2.3 Impact and Stakeholder Engagement

The goal of an operational quality study with these adjustments is to build models that are not only accurate but also actionable. Successfully modeling wildfire activity can significantly enhance transparency, support critical decision-making, and improve communication with stakeholders. By overcoming the "black box" reputation of DL models through the implementation of feature importance plots (SHAP) and providing interpretable confidence intervals from probabilistic models, we can empower users to better understand how to leverage model outputs. Land managers, for instance, could utilize risk heatmaps for risk prioritization, optimize resource allocation, and communicate potential warnings more effectively to the public. Ultimately, this work demonstrates a comprehensive deep learning workflow capable of addressing complex scientific questions, contributing to a deeper understanding of real-world phenomena, and ultimately informing impactful decisions for a safer world.

6. Conclusion

This chapter provides guidance on how to use Bayesian Neural Networks (BNNs) to classify satellite-detected active fire hotspots and quantify the uncertainty associated with each hotspot. We demonstrate that Probabilistic BNNs can effectively provide more contextual information about active fire hotspot data by leveraging uncertainty estimates, with improved performance when using balanced datasets and hyperparameter tuning. This type of study could be improved by incorporating multi-temporal data, expanding spatial scale, and exploring alternative data balancing methods, and applying the approach to additional sensors.

Additionally, we could integrate more uncertainty information into near-real-time decision-making workflows to provide additional context about hotspot confidences and multi-source agreement.

7. Acknowledgements

We would like to thank Peter Engelfield (Northern Forestry Centre, Natural Resource Canada) for providing the persistent hotspots dataset, which was essential to this work. We are also grateful to Piyush Jain (Northern Forestry Centre, Natural Resources Canada) for his thoughtful review and feedback on the manuscript. Finally, we thank the GeoAI Working Group for the opportunity to contribute to their textbook and for fostering a collaborative and supportive research community.

8. References

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., ... Zheng, X. (2016). *TensorFlow: Large-Scale Machine Learning on Heterogeneous Distributed Systems* (No. arXiv:1603.04467). arXiv. <https://doi.org/10.48550/arXiv.1603.04467>
- Alencar, A. A. C., Arruda, V. L. S., Silva, W. V. da, Conciani, D. E., Costa, D. P., Crusco, N., Duverger, S. G., Ferreira, N. C., Franca-Rocha, W., Hasenack, H., Martenexen, L. F. M., Piontekowski, V. J., Ribeiro, N. V., Rosa, E. R., Rosa, M. R., dos Santos, S. M. B., Shimbo, J. Z., & Vélez-Martin, E. (2022). Long-Term Landsat-Based Monthly Burned Area Dataset for the Brazilian Biomes Using Deep Learning. *Remote Sensing*, 14(11), Article 11. <https://doi.org/10.3390/rs14112510>
- Arbel, J., Pitas, K., Vladimirova, M., & Fortuin, V. (2023). *A Primer on Bayesian Neural Networks: Review and Debates* (No. arXiv:2309.16314). arXiv. <https://doi.org/10.48550/arXiv.2309.16314>
- Ba, R., Chen, C., Yuan, J., Song, W., & Lo, S. (2019). SmokeNet: Satellite Smoke Scene Detection Using Convolutional Neural Network with Spatial and Channel-Wise Attention. *Remote Sensing*, 11(14), Article 14. <https://doi.org/10.3390/rs11141702>
- Ban, Y., Zhang, P., Nascetti, A., Bevington, A. R., & Wulder, M. A. (2020). Near Real-Time Wildfire Progression Monitoring with Sentinel-1 SAR Time Series and Deep Learning.

- Scientific Reports*, 10(1), 1322. <https://doi.org/10.1038/s41598-019-56967-x>
- Beverly, J. L., & Schroeder, D. (2025). Alberta's 2023 wildfires: Context, factors, and futures. *Canadian Journal of Forest Research*, 55, 1–19. <https://doi.org/10.1139/cjfr-2024-0099>
- Blundell, C., Cornebise, J., Kavukcuoglu, K., & Wierstra, D. (2015). Weight Uncertainty in Neural Network. *Proceedings of the 32nd International Conference on Machine Learning*, 1613–1622. <https://proceedings.mlr.press/v37/blundell15.html>
- Boulanger, Y., Arseneault, D., Bélisle, A. C., Bergeron, Y., Boucher, J., Boucher, Y., Danneyrolles, V., Erni, S., Gachon, P., Girardin, M. P., Grant, E., Grondin, P., Jetté, J.-P., Labadie, G., Leblond, M., Leduc, A., Puigdevall, J. P., St-Laurent, M.-H., Tremblay, J. A., & Waldron, K. (2025). The 2023 wildfire season in Québec: An overview of extreme conditions, impacts, lessons learned, and considerations for the future. *Canadian Journal of Forest Research*, 55, 1–21. <https://doi.org/10.1139/cjfr-2023-0298>
- Canada, N. R. (n.d.). *Canadian Wildland Fire Information System | Fire M3 Hotspots*. Retrieved May 27, 2025, from <https://cwfis.cfs.nrcan.gc.ca/maps/fm3?type=tri>
- Chen, Y., Morton, D. C., & Randerson, J. T. (2024). Remote sensing for wildfire monitoring: Insights into burned area, emissions, and fire dynamics. *One Earth*, 7(6), 1022–1028. <https://doi.org/10.1016/j.oneear.2024.05.014>
- Cho, A. Y., Park, S., Kim, D., Kim, J., Li, C., & Song, J. (2023). Burned Area Mapping Using Unitemporal PlanetScope Imagery With a Deep Learning Based Approach. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16, 242–253. <https://doi.org/10.1109/JSTARS.2022.3225070>
- Chuvieco, E., Aguado, I., Salas, J., García, M., Yebra, M., & Oliva, P. (2020). Satellite Remote Sensing Contributions to Wildland Fire Science and Management. *Current Forestry Reports*, 6(2), 81–96. <https://doi.org/10.1007/s40725-020-00116-5>
- Chuvieco, E., Mouillot, F., van der Werf, G. R., San Miguel, J., Tanase, M., Koutsias, N., García, M., Yebra, M., Padilla, M., Gitas, I., Heil, A., Hawbaker, T. J., & Giglio, L. (2019).

- Historical background and current developments for mapping burned area from satellite Earth observation. *Remote Sensing of Environment*, 225, 45–64.
<https://doi.org/10.1016/j.rse.2019.02.013>
- Coogan, S. C. P., Daniels, L. D., Boychuk, D., Burton, P. J., Flannigan, M. D., Gauthier, S., Kafka, V., Park, J. S., & Wotton, B. M. (2021). Fifty years of wildland fire science in Canada. *Canadian Journal of Forest Research*, 51(2), 283–302.
<https://doi.org/10.1139/cjfr-2020-0314>
- Crowley, M. A., Hope, E. S., McFayden, C. B., MacPherson, L., Brehaut, L., de Jong, M. C., Beckers, J. F., Dufour, D., Harvie, J., Goetz, G., DeBoer, S., Ellicott, E., Shuman, J. K., Smith, M., Morton, D. C., Cole, T., van Mierlo, H., Paul, N., Wiggins, B., ... Johnston, J. M. (In Review). *The importance of engaging with a variety of users for collaborative development in end-to-end satellite missions: A case study from the WildFireSat mission*.
- Crowley, M. A., MacPherson, L., McFayden, C. B., Johnston, J. M., Dufour, D., DeBoer, S., Mizzi, S., Hope, E. S., de Jong, M. C., Thompson, D. K., & Cantin, A. S. (2023b). *Forming a constellation of connections: The 2023 WildFireSat stakeholders meeting*. (Information Report No. GLC-WF-1).
<https://ostrnrcan-dostrncan.canada.ca/entities/publication/acc746a8-78f6-48e7-8496-67ac07d773f6?fromSearchPage=true>
- Crowley, M. A., Stockdale, C. A., Johnston, J. M., Wulder, M. A., Liu, T., McCarty, J. L., Rieb, J. T., Cardille, J. A., & White, J. C. (2023). Towards a whole-system framework for wildfire monitoring using Earth observations. *Global Change Biology*, 29(6), 1423–1436.
<https://doi.org/10.1111/gcb.16567>
- Cunningham, C. X., Williamson, G. J., & Bowman, D. M. J. S. (2024). Increasing frequency and intensity of the most extreme wildfires on Earth. *Nature Ecology & Evolution*, 8(8), 1420–1425. <https://doi.org/10.1038/s41559-024-02452-2>

- Daniels, L. D., Dickson-Hoyle, S., Baron, J. N., Copes-Gerbitz, K., Flannigan, M. D., Castellanos-Acuna, D., Hoffman, K. M., Bourbonnais, M., Wilkinson, S. L., Roeser, D., Harvey, J. E., Laflamme, J., Tiribelli, F., Whitehead, J., Leverkus, S. E. R., & Gray, R. W. (2024). The 2023 wildfires in British Columbia, Canada: Impacts, drivers, and transformations to coexist with wildfire. *Canadian Journal of Forest Research*.
<https://doi.org/10.1139/cjfr-2024-0092>
- Davies, D. (2025, March 7). *FIRMS Releases New Features to Identify Active Fires by Type* | *NASA Earthdata* [Blog]. Earth Science Data Systems, NASA.
<https://www.earthdata.nasa.gov/news/blog/firms-releases-new-features-identify-active-fires-type>
- Davies, D. K., Murphy, K. J., Michael, K., Becker-Reshef, I., Justice, C. O., Boller, R., Braun, S. A., Schmaltz, J. E., Wong, M. M., Pasch, A. N., Dye, T. S., da Silva, A. M., Goodman, H. M., & Morin, P. J. (2015). The Use of NASA LANCE Imagery and Data for Near Real-Time Applications. In C. D. Lippitt, D. A. Stow, & L. L. Coulter (Eds.), *Time-Sensitive Remote Sensing* (pp. 165–182). Springer.
https://doi.org/10.1007/978-1-4939-2602-2_11
- de Jong, M. C., & McFayden, C. B. (2023). *Wildfires from Space: The end of MODIS fire data*. *Frontline Express* 92. <https://ostrnrcan-dostrnrcan.canada.ca/handle/1845/218131>
- Earth Science Data Systems, N. (2024a, November 1). *FIRMS FAQ* | *NASA Earthdata* [Basic page]. Earth Science Data Systems, NASA.
<https://www.earthdata.nasa.gov/data/tools/firms/faq>
- Earth Science Data Systems, N. (2024b, November 1). *FIRMS FAQ* | *NASA Earthdata* [Basic page]. Earth Science Data Systems, NASA.
<https://www.earthdata.nasa.gov/data/tools/firms/faq>
- Fraser, R. H., Li, Z., & Cihlar, J. (2000). Hotspot and NDVI Differencing Synergy (HANDS): A New Technique for Burned Area Mapping over Boreal Forest. *Remote Sensing of*

- Environment*, 74(3), 362–376. [https://doi.org/10.1016/S0034-4257\(00\)00078-X](https://doi.org/10.1016/S0034-4257(00)00078-X)
- Fu, Y., Li, R., Wang, X., Bergeron, Y., Valeria, O., Chavardès, R. D., Wang, Y., & Hu, J. (2020). Fire Detection and Fire Radiative Power in Forests and Low-Biomass Lands in Northeast Asia: MODIS versus VIIRS Fire Products. *Remote Sensing*, 12(18), Article 18. <https://doi.org/10.3390/rs12182870>
- Ghali, R., & Akhloufi, M. A. (2023). BoucaNet: A CNN-Transformer for Smoke Recognition on Remote Sensing Satellite Images. *Fire*, 6(12), Article 12. <https://doi.org/10.3390/fire6120455>
- Giglio, L., Schroeder, W., & Justice, C. O. (2016a). The collection 6 MODIS active fire detection algorithm and fire products. *Remote Sensing of Environment*, 178, 31–41. <https://doi.org/10.1016/j.rse.2016.02.054>
- Girardin, M. P., Ali, A. A., & Hély, C. (2010). Wildfires in boreal ecosystems: Past, present and some emerging trends. *International Journal of Wildland Fire*, 19(8), 991–995. https://doi.org/10.1071/WFv19n8_FO
- Girardin, M. P., & Mudelsee, M. (2008). Past and Future Changes in Canadian Boreal Wildfire Activity. *Ecological Applications*, 18(2), 391–406. <https://doi.org/10.1890/07-0747.1>
- Gonçalves, L. A. O., Ghali, R., & Akhloufi, M. A. (2024). YOLO-Based Models for Smoke and Wildfire Detection in Ground and Aerial Images. *Fire*, 7(4), Article 4. <https://doi.org/10.3390/fire7040140>
- Google. (2025). *Google Colab*. <https://colab.research.google.com/>
- Hope, E. S., McFayden, C. B., Jong, M. C. de, Crowley, M. A., Johnston, J. M., Davies, D. K., Olsina, O., Hewson, J., San-Miguel-Ayanz, J., Ferrari, D., & Sloane, M. (2024). Exploring the use of satellite Earth observation active wildland fire hotspot data via open access web platforms. *International Journal of Digital Earth*. <https://www.tandfonline.com/doi/abs/10.1080/17538947.2024.2420821>
- Hu, X., Ban, Y., & Nascetti, A. (2021). Uni-Temporal Multispectral Imagery for Burned Area

Mapping with Deep Learning. *Remote Sensing*, 13(8), Article 8.

<https://doi.org/10.3390/rs13081509>

Jain, P., Barber, Q. E., Taylor, S. W., Whitman, E., Castellanos Acuna, D., Boulanger, Y., Chavardès, R. D., Chen, J., Englefield, P., Flannigan, M., Girardin, M. P., Hanes, C. C., Little, J., Morrison, K., Skakun, R. S., Thompson, D. K., Wang, X., & Parisien, M.-A. (2024). Drivers and Impacts of the Record-Breaking 2023 Wildfire Season in Canada.

Nature Communications, 15(1), 6764. <https://doi.org/10.1038/s41467-024-51154-7>

Jain, P., Coogan, S. C. P., Subramanian, S. G., Crowley, M., Taylor, S., & Flannigan, M. D.

(2020). A review of machine learning applications in wildfire science and management.

Environmental Reviews, 28(4), 478–505. <https://doi.org/10.1139/er-2020-0019>

Johnston, J. M., Jackson, N., McFayden, C., Ngo Phong, L., Lawrence, B., Davignon, D.,

Wooster, M. J., van Mierlo, H., Thompson, D. K., Cantin, A. S., Johnston, D., Johnston,

L. M., Sloane, M., Ramos, R., & Lynham, T. J. (2020). Development of the User

Requirements for the Canadian WildFireSat Satellite Mission. *Sensors*, 20(18), Article

18. <https://doi.org/10.3390/s20185081>

Johnston, J. M., Johnston, L. M., Wooster, M. J., Brookes, A., McFayden, C., & Cantin, A. S.

(2018). Satellite Detection Limitations of Sub-Canopy Smouldering Wildfires in the North

American Boreal Forest. *Fire*, 1(2), Article 2. <https://doi.org/10.3390/fire1020028>

Jones, M. W., Abatzoglou, J. T., Veraverbeke, S., Andela, N., Lasslop, G., Forkel, M., Smith, A.

J. P., Burton, C., Betts, R. A., van der Werf, G. R., Sitch, S., Canadell, J. G., Santín, C.,

Kolden, C., Doerr, S. H., & Le Quéré, C. (2022). Global and Regional Trends and Drivers of Fire Under Climate Change. *Reviews of Geophysics*, 60(3), e2020RG000726.

<https://doi.org/10.1029/2020RG000726>

Keras. (2021). *Keras documentation: Probabilistic Bayesian Neural Networks*.

https://keras.io/examples/keras_recipes/bayesian_neural_networks/

Kirchmeier-Young, M. C., Malinina, E., Barber, Q. E., Garcia Perdomo, K., Curasi, S. R., Liang,

- Y., Jain, P., Gillett, N. P., Parisien, M.-A., Cannon, A. J., Lima, A. R., Arora, V. K., Boulanger, Y., Melton, J. R., Van Vliet, L., & Zhang, X. (2024). Human driven climate change increased the likelihood of the 2023 record area burned in Canada. *Npj Climate and Atmospheric Science*, 7(1), 1–12. <https://doi.org/10.1038/s41612-024-00841-9>
- Knopp, L., Wieland, M., Rättich, M., & Martinis, S. (2020). A Deep Learning Approach for Burned Area Segmentation with Sentinel-2 Data. *Remote Sensing*, 12(15), Article 15. <https://doi.org/10.3390/rs12152422>
- Kufel, J., Bargieł-Łączek, K., Kocot, S., Koźlik, M., Bartnikowska, W., Janik, M., Czogalik, Ł., Dudek, P., Magiera, M., Lis, A., Paszkiewicz, I., Nawrat, Z., Cebula, M., & Gruszczyńska, K. (2023). What Is Machine Learning, Artificial Neural Networks and Deep Learning?—Examples of Practical Applications in Medicine. *Diagnostics*, 13(15), Article 15. <https://doi.org/10.3390/diagnostics13152582>
- Larsen, A., Hanigan, I., Reich, B. J., Qin, Y., Cope, M., Morgan, G., & Rappold, A. G. (2021). A deep learning approach to identify smoke plumes in satellite imagery in near-real time for health risk communication. *Journal of Exposure Science & Environmental Epidemiology*, 31(1), 170–176. <https://doi.org/10.1038/s41370-020-0246-y>
- Lee, K., Kim, J., Chong, S., & Shin, J. (2016). *Making Stochastic Neural Networks from Deterministic Ones*. <https://openreview.net/forum?id=B1akgy9xx>
- Liu, J., Li, J., Peters, S., & Zhao, L. (2024). A transformer boosted UNet for smoke segmentation in complex backgrounds in multispectral LandSat imagery. *Remote Sensing Applications: Society and Environment*, 36, 101283. <https://doi.org/10.1016/j.rsase.2024.101283>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 30). Curran Associates, Inc.

https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf

Ma, Y., Zhang, Z., Kang, Y., & Özdoğan, M. (2021). Corn yield prediction and uncertainty analysis based on remotely sensed variables using a Bayesian neural network approach. *Remote Sensing of Environment*, 259, 112408.
<https://doi.org/10.1016/j.rse.2021.112408>

McFayden, C. B., Hope, E. S., Boychuk, D., Johnston, L. M., Richardson, A., Coyle, M., Sloane, M., Cantin, A. S., Johnston, J. M., & Lynham, T. J. (2023a). Canadian Fire Management Agency Readiness for WildFireSat: Assessment and Strategies for Enhanced Preparedness. *Fire*, 6(2), Article 2. <https://doi.org/10.3390/fire6020073>

McFayden, C. B., Hope, E. S., Johnston, J. M., Cantin, A. S., Crowley, M. A., DeBoer, S., de Jong, M. C., Johnston, L. M., Thompson, D. K., & Young, D. J. (2023b). *WildFireSat pathway for implementation and uptake in provincial and territorial fire management agencies*. (Information Report No. GLC-X-36; p. 45).
<https://ostrnrcan-dostrncan.canada.ca/entities/publication/648a6f60-d555-4171-99e7-86ad74ddeab8>

McFayden, C. B., Johnston, J. M., Brehaut, L., Cantin, A. S., Chaffey, A., Crowley, M. A., de Jong, M. C., Diaz, E., Lecours, M. J., Ly, V., McAlpine, R., Motty, J., & Thompson, D. K. (2025). *Charting Constellations: The 2024 CIFFC and WildFireSat Fire Agency Meeting* (pp. 1-26.) [Information Report GLC-WF-3. Information Report (GLFC - Sault Ste. Marie), 3, 1-26.]. Great Lakes Forestry Centre.

Mullachery, V., Khera, A., & Husain, A. (2018). *Bayesian Neural Networks* (No. arXiv:1801.07710). arXiv. <https://doi.org/10.48550/arXiv.1801.07710>

Natural Resources Canada. (2025a). *Canadian National Fire Database* [Dataset].
<https://cwfis.cfs.nrcan.gc.ca/ha/nfdb?type=poly&year=9999>

Natural Resources Canada. (2025b). *National Burned Area Composite* [Dataset].

- <https://cwfis.cfs.nrcan.gc.ca/ha/nfdb?type=nbac&year=9999>
- Nguyen, H. A. D., Ha, Q. P., Duc, H., Azzi, M., Jiang, N., Barthelemy, X., & Riley, M. (2023). Long Short-Term Memory Bayesian Neural Network for Air Pollution Forecast. *IEEE Access*, 11, 35710–35725. <https://doi.org/10.1109/ACCESS.2023.3265725>
- Olofsson, P., Foody, G. M., Herold, M., Stehman, S. V., Woodcock, C. E., & Wulder, M. A. (2014). Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment*, 148, 42–57. <https://doi.org/10.1016/j.rse.2014.02.015>
- O’Shea, K., & Nash, R. (2015). *An Introduction to Convolutional Neural Networks* (No. arXiv:1511.08458). arXiv. <https://doi.org/10.48550/arXiv.1511.08458>
- Panchal, G., Ganatra, A., Kosta, Y. P., & Panchal, D. (2011). Behaviour Analysis of Multilayer Perceptrons with Multiple Hidden Neurons and Hidden Layers. *International Journal of Computer Theory and Engineering*, 332–337. <https://doi.org/10.7763/IJCTE.2011.V3.328>
- Plotly Technologies Inc. (2015). *Collaborative data science*. Plotly Technologies Inc. <https://plot.ly>
- Python. (2021). *Python Release Python 3.10.0*. Python.Org. <https://www.python.org/downloads/release/python-3100/>
- Russell, S. J., & Norvig, P. (with Chang, M., Devlin, J., Dragan, A., Forsyth, D., Goodfellow, I., Malik, J., Mansinghka, V., Pearl, J., & Wooldridge, M. J.). (2022). *Artificial intelligence: A modern approach* (Fourth edition, global edition). Pearson.
- San-Miguel-Ayanz, J., & Ravail, N. (2005). Active Fire Detection for Fire Emergency Management: Potential and Limitations for the Operational Use of Remote Sensing. *Natural Hazards*, 35(3), 361–376. <https://doi.org/10.1007/s11069-004-1797-2>
- Schmidt, R. M. (2019). *Recurrent Neural Networks (RNNs): A gentle Introduction and Overview* (No. arXiv:1912.05911). arXiv. <https://doi.org/10.48550/arXiv.1912.05911>
- Schroeder, W., Oliva, P., Giglio, L., & Csiszar, I. A. (2014). The New VIIRS 375 m active fire detection data product: Algorithm description and initial assessment. *Remote Sensing of*

- Environment*, 143, 85–96. <https://doi.org/10.1016/j.rse.2013.12.008>
- Seydi, S. T., Hasanlou, M., & Chanussot, J. (2022). *Burnt-Net*: Wildfire burned area mapping with single post-fire Sentinel-2 data and deep learning morphological neural network. *Ecological Indicators*, 140, 108999. <https://doi.org/10.1016/j.ecolind.2022.108999>
- Shlens, J. (2014). *Notes on Kullback-Leibler Divergence and Likelihood* (No. arXiv:1404.2000). arXiv. <https://doi.org/10.48550/arXiv.1404.2000>
- Skakun, R., Castilla, G., & Jain, P. (2024). Mapping wildfires in Canada with Landsat MSS to extend the National Burned Area Composite (NBAC) time series back to 1972. *International Journal of Wildland Fire*, 33(12). <https://doi.org/10.1071/WF24138>
- Skakun, R., Castilla, G., Metsaranta, J., Whitman, E., Rodrigue, S., Little, J., Groenewegen, K., & Coyle, M. (2022). Extending the National Burned Area Composite Time Series of Wildfires in Canada. *Remote Sensing*, 14(13), Article 13. <https://doi.org/10.3390/rs14133050>
- Tensorflow. (2025). *TensorFlow Probability*. <https://www.tensorflow.org/probability>
- Thompson, D. K., de Jong, M. C., Crowley, M. A., Ansell, M., Cantin, A. S., Goetz, G., Harvie, J., MacPherson, L., McFayden, C. B., Oladipo, S., Oliver, J., Sloane, M., & Johnston, J. M. (2025). *Wildfire Science and Applications Plan*. Natural Resources Canada. <https://ostrnrcan-dostrncan.canada.ca/handle/1845/307362>
- Ting, K. M. (2017). Confusion Matrix. In C. Sammut & G. I. Webb (Eds.), *Encyclopedia of Machine Learning and Data Mining* (pp. 260–260). Springer US. https://doi.org/10.1007/978-1-4899-7687-1_50
- Tymstra, C., Stocks, B. J., Cai, X., & Flannigan, M. D. (2020). Wildfire management in Canada: Review, challenges and opportunities. *Progress in Disaster Science*, 5, 100045. <https://doi.org/10.1016/j.pdisas.2019.100045>
- Wang, W., Wang, X., Flannigan, M. D., Guindon, L., Swystun, T., Castellanos-Acuna, D., Wu, W., & Wang, G. (2025). Canadian forests are more conducive to high-severity fires in

- recent decades. *Science*, 387(6729), 91–97. <https://doi.org/10.1126/science.ado1006>
- Wang, X., Parisien, M.-A., Taylor, S. W., Candau, J.-N., Stralberg, D., Marshall, G. A., Little, J. M., & Flannigan, M. D. (2017). Projected changes in daily fire spread across Canada over the next century. *Environmental Research Letters*, 12(2), 025005.
<https://doi.org/10.1088/1748-9326/aa5835>
- Wooster, M. J., Roberts, G. J., Giglio, L., Roy, D. P., Freeborn, P. H., Boschetti, L., Justice, C., Ichoku, C., Schroeder, W., Davies, D., Smith, A. M. S., Setzer, A., Csiszar, I., Strydom, T., Frost, P., Zhang, T., Xu, W., de Jong, M. C., Johnston, J. M., ... San-Miguel-Ayanz, J. (2021). Satellite remote sensing of active fires: History and current status, applications and future requirements. *Remote Sensing of Environment*, 267, 112694.
<https://doi.org/10.1016/j.rse.2021.112694>
- Wu, Y., & Feng, J. (2018). Development and Application of Artificial Neural Network. *Wireless Personal Communications*, 102(2), 1645–1656.
<https://doi.org/10.1007/s11277-017-5224-x>
- Zhao, Y., & Ban, Y. (2025). Assessment of L-band and C-band SAR on Burned Area Mapping of Multi-Severity Forest Fires using Deep Learning. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 1–12.
<https://doi.org/10.1109/JSTARS.2025.3560287>